

Hotel Reservation System

Otrip



System Analysis & Design Capstone Project

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Introduction

Imagine a world where managing a hotel is as simple as tapping a few buttons on your smartphone. Our hotel management system (OTrip) is proposed with a view to ensure the overall success and growth of the hotel industry by addressing the challenges faced by both hotel staff and customers. By leveraging modern technology, OTrip will provide a user-friendly interface for hotel staff to manage tasks efficiently, enhance customer experiences, and ultimately improve overall hotel performance. Centralised platform for managing hotel operations, including reservations, check-ins/check-outs, room assignments, and customer information and many more functionalities.

Objective

In order to guarantee a flawless user experience and effective management for hotel staff, a number of integrations and functionalities are necessary to design a full hotel reservation application. With its extensive web-based platform, the Hotel Reservation Application (OTrip) seeks to make reservations easier for both customers and hotel employees. The app will collect and analyse data to optimise operations and make informed decisions. Additionally, it will focus on cost reduction, revenue generation, scalability, security, and user-friendliness to deliver a comprehensive and valuable solution for hotels. Furthermore, Otrip will integrate with existing hotel systems and offer customizable features to meet the specific needs of different hotel properties.

Key Features

- **Centralised Dashboard:** A thorough overview of hotel operations, including reservations, room status, customer information and previous Customer reviews.
- **User Registration and login:** The system makes it easy for users to register, and its strong authentication procedures guarantee safe access with password recovery options.
- **Reservation Management:** Seamless booking and cancellation processes, along with customer information management.
- **Check-in/Check-out:** Efficient check-in and check-out procedures, including digital signature and payment processing.
- **Payment System:** Easy payment system with multiple payment methods.

- **Room Management:** Real-time room status updates, cleaning assignments, and maintenance requests.
- **Customer Profile Management:** Store and manage customer profiles for personalised services and future stays.
- **In-App Messaging:** Enable customers to communicate directly with hotel staff for requests and inquiries.
- **Review System:** Customers will be able to see reviews and put their own.

Scenario Writing

Scenario 1: Profile Management

A customer wants to view or update their profile information.

- Logs into their account and navigates to their profile settings.
- Displays the customer's personal information, preferences, and booking history.
- Makes any necessary changes and saves their updates.
- Updates the customer's profile information.

Scenario 2: Room Reservation

A customer wants to book a room at a hotel.

- Available rooms based on desired dates, location, and preferences.
- Displays available rooms, rates, and amenities.
- Selects a room and proceeds to the booking process.
- Collects customer information, payment details, and confirmation preferences.
- Completes the booking process and receives a confirmation email or SMS.
- Updates the hotel's reservation system with the new booking.

Scenario 3: Check-In

A customer arrives at the hotel for their check-in.

- Presents their booking confirmation or identification.
- Verifies the booking and customer information.

- Displays the customer's reservation details on the staff's device.
- Processes the check-in, including room assignment and payment.
- Signs a digital check-in form and receives their room key.
- Updates the room status to "Occupied" and records the check-in time.

Scenario 4: Check-Out

A customer is checking out of the hotel.

- Presents their room key.
- Verifies the customer's identity and checks for any outstanding charges.
- Calculates the final bill and processes any payments.
- Signs a digital check-out form and returns their room key.
- Updates the room status to "Vacant" and records the check-out time.

Scenario 5: Payment System

A customer wants to pay for their reservation.

- Selects their preferred payment method (e.g., credit card, debit card, digital wallet).
- Redirects the customer to a secure payment gateway.
- Enters their payment information and completes the transaction.
- Verifies the payment and updates the reservation status.
- Receives a payment confirmation.

These scenarios provide practical examples of how different users interact with the Hotel Management System and how the system responds to their actions. Scenarios like these help stakeholders understand the system's behaviour and can be used for testing, validation, and refining the software to meet user needs effectively.

Stakeholder

The key stakeholders involved in this Hotel Management System are -

- Administrator
- Manager
- Customer

User Profile

User profile 1 – Administrator

User Class	Notes on Characteristics	Requirement Implied
Types of Users	Administrator	Verification
Age Range	30-40	Verification
Frequency of Use	Daily	Performance, Operation, Acceptance
Mandatory	yes	
Computer Experience	Experienced with system administration	Documentation
Education	M. Sc	Documentation
Goals	Ensure the system is user-friendly and safe	Performance, Operation, Acceptance, Resource, Security
Language skills	Bangla, English	
Number of Users	12-20	Performance, Operation, Portability
Training	If needed	Documentation
Other System Used	No	
Ways of Working	Worked for maintaining the system	Acceptance, Maintenance, Portability, Safety, Security, Operation

User profile 2 - Manager

User Class	Notes on Characteristics	Requirement Implied
Types of Users	Manager	Verification
Age Range	25-40	Verification
Frequency of Use	Daily	Performance, Operation, Acceptance
Mandatory	Yes	
Computer Experience	Experienced with Hotel Reservation System	Documentation
Education	Graduate	Documentation
Goals	Ensure quality service, maintain security	Performance, Operation, Acceptance, Resource, Security
Language skills	Bangla, English	
Number of Users	20-25	Performance, Operation, Portability
Training	Initial training if needed	Documentation
Other System Used	No	
Ways of Working	Worked for maintaining the system	Acceptance, Maintenance, Portability, Safety, Security, Operation

User profile 3 - Customer

User Class	Notes on Characteristics	Requirement Implied
Types of Users	Customer	Verification
Age Range	Any age above 18	Verification
Frequency of Use	When they want to	Performance, Operation, Acceptance
Mandatory	No	
Computer Experience	No	
Goals	Access quality service	Performance, Operation, Acceptance, Resource, Security
Language skills	Bangla, English	
Number of Users	70-120	Acceptance, Portability
Training	No specific training	
Other System Used	No	

Scope

Centralised Dashboard

- **Objective:** Provide a comprehensive overview of hotel operations to improve efficiency and decision-making.
- **Functionality:** Real-time data visualisation, customizable dashboards, integration with other hotel systems (e.g., PMS, CRM), historical data analysis.

User Registration and Login

- **Objective:** Enable secure and convenient access to the system for users.
- **Functionality:** Strong authentication mechanisms (e.g., password hashing, two-factor authentication), password recovery options, user role management.

Reservation Management

- **Objective:** Streamline the booking process and manage customer information effectively.
- **Functionality:** Online booking engine, group booking capabilities, automated confirmation emails, integration with payment gateways, cancellation policies.

Check-In/Check-Out

- **Objective:** Efficiently handle customer arrivals and departures, ensuring a seamless experience.
- **Functionality:** Digital check-in/check-out options, automated room assignments, integration with property management systems, real-time updates on room status.

Payment System

- **Objective:** Facilitate secure and convenient payment processing for customers.
- **Functionality:** Multiple payment methods (e.g., credit card, debit card, digital wallets), secure payment processing, integration with accounting software, automated invoicing.

Room Management

- **Objective:** Optimise room utilisation, manage housekeeping tasks, and track maintenance requests.
- **Functionality:** Real-time room status updates, housekeeping task management, maintenance request tracking, room inventory management.

Customer Profile Management

- **Objective:** Store and manage customer information to provide personalised services and enhance customer loyalty.
- **Functionality:** Customer preferences (e.g., room type, amenities), booking history, loyalty program integration, communication preferences.

In-App Messaging

- **Objective:** Enable efficient communication between Customers and hotel staff for requests and inquiries.
- **Functionality:** Real-time messaging, message history, integration with other hotel systems (e.g., PMS), automated notifications.

Review System

- **Objective:** Collect and manage customer feedback to improve hotel services and reputation.
- **Functionality:** Review moderation, integration with reputation management tools, automated review requests, response management.

Feasibility Study

This feasibility study evaluates the proposed Hotel Reservation System (Otrip) by examining its technical, financial, market, and organisational aspects. The goal is to determine the viability of the project and its potential for successful implementation.

1. Technical Feasibility

A. Hardware Compatibility: The system must be compatible with existing hardware infrastructure in hotels, such as computers, servers, and networking equipment. A review of current hardware will be conducted to identify any necessary upgrades.

B. Software Compatibility: The Otrip should integrate seamlessly with existing software systems used by hotels, including property management systems (PMS), accounting software, and customer relationship management (CRM) tools. Compatibility with various operating systems and databases will be evaluated.

C. Technical Expertise: The project will assess the availability of skilled IT personnel to develop, implement, and maintain the system.

D. Security Measures: The system will implement industry-standard security protocols, such as SSL encryption, secure user authentication, and regular updates to protect against vulnerabilities.

2. Financial Feasibility

A. Cost-Benefit Analysis

Development Costs: This includes expenses for software development, hardware upgrades, and training programs.

Operational Costs: Ongoing costs such as maintenance, support, and hosting will be estimated.

Expected Benefits: Increased efficiency, enhanced customer satisfaction, and potential revenue growth from improved booking processes will be quantified.

B. Return on Investment (ROI): The projected ROI will be calculated over a defined period (e.g., 3-5 years) by comparing the total costs against the expected financial benefits. This will help stakeholders understand the financial viability of the project.

3. Market Feasibility

A. Target Market Analysis

Market Demand: A study will be carried out to evaluate the need for hotel reservation systems in the hospitality sector. Analysing consumer preferences and internet booking trends is part of this.

Competitive Analysis: An evaluation of existing competitors in the market will identify strengths, weaknesses, and unique selling propositions for the proposed system.

B. User Acceptance: Surveys and focus groups with potential users will be conducted to gather insights on their needs and expectations, which will inform system design and functionality.

4. Organizational Feasibility

A. Stakeholder Involvement: In order to meet their needs and promote approval, stakeholders will be involved at every stage of the development process.

B. Impact on Current Operations: The study will analyse how the implementation of the Otrip will affect existing workflows and processes. Strategies will be developed to minimise disruptions during the transition.

C. Training and Support: Comprehensive training programs will be developed for all user groups to ensure smooth adoption of the new system.

Conclusion

According to the feasibility assessment, the Hotel Reservation System makes sense from an organisational, financial, and technical aspect. The expected advantages, such as better customer satisfaction, increased operational efficiency, and increased potential revenue, exceed the related expenses. To guarantee successful execution, strategies for optimising benefits and recommendations for risk reduction will be integrated into the project plan.

Block Diagram

Block Diagram-1 : Customer

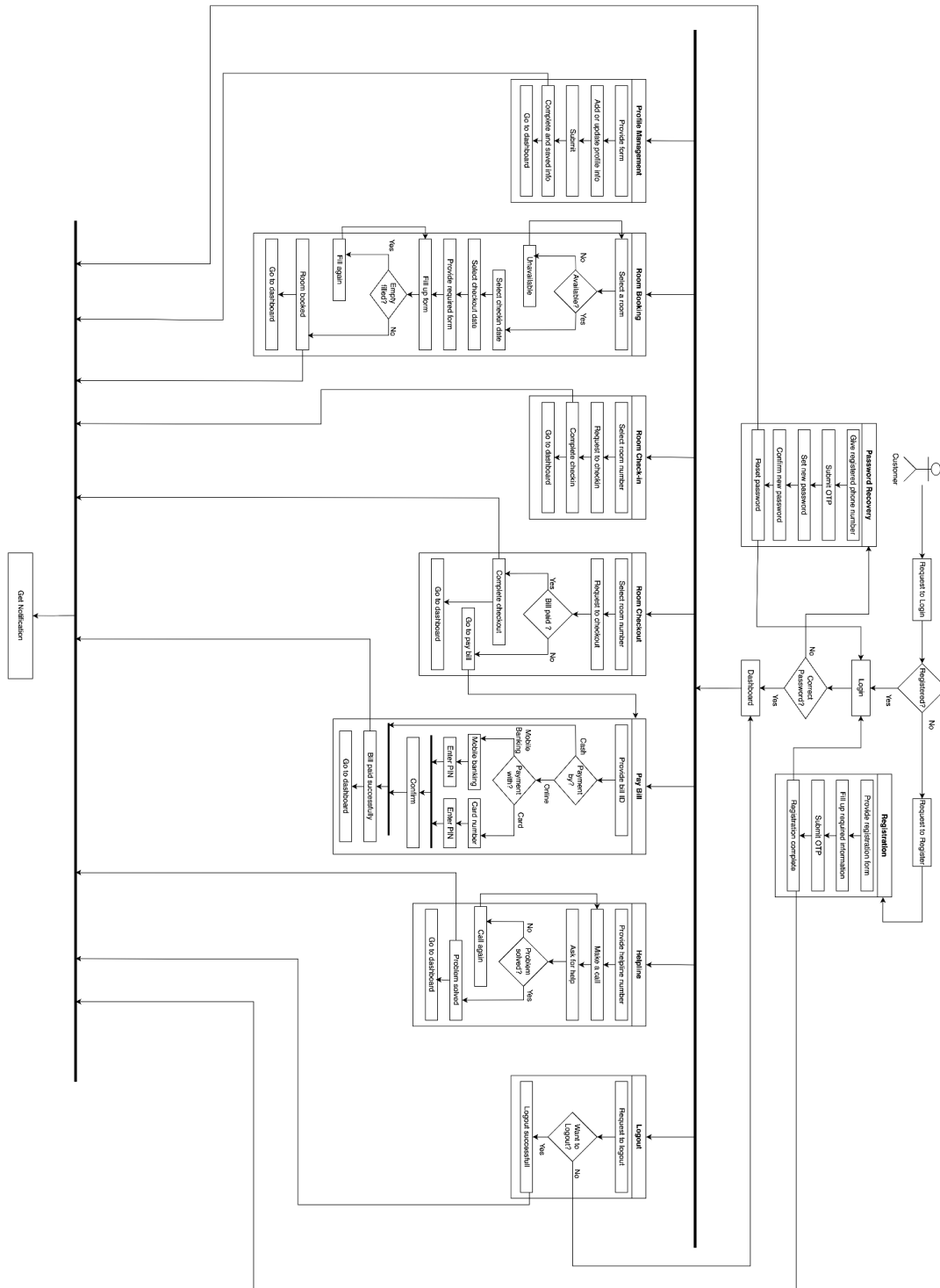


Figure: Block diagram for Customer

Block Diagram-2: Manager

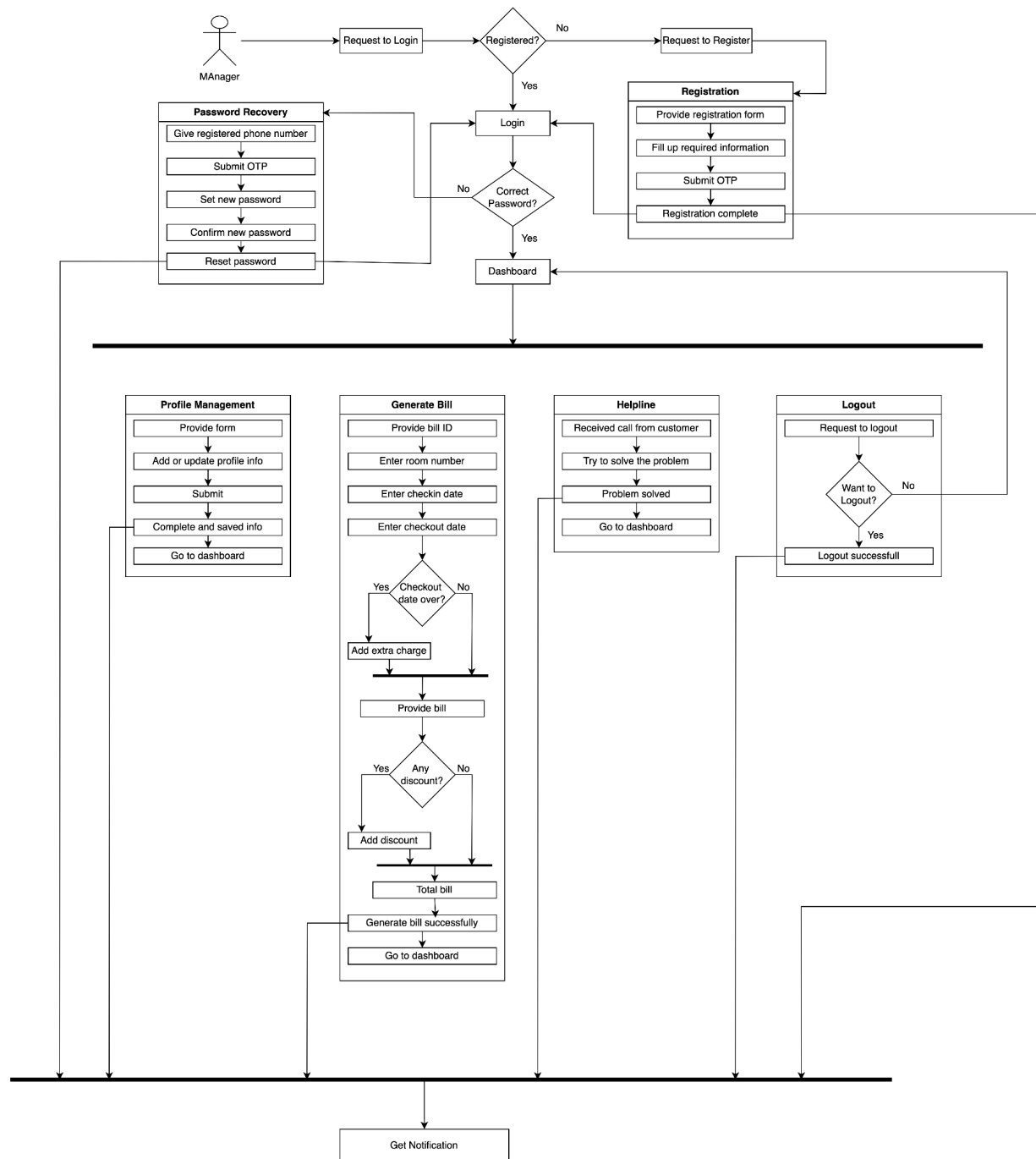


Figure: Block diagram for manager

Block Diagram-3: Administrator

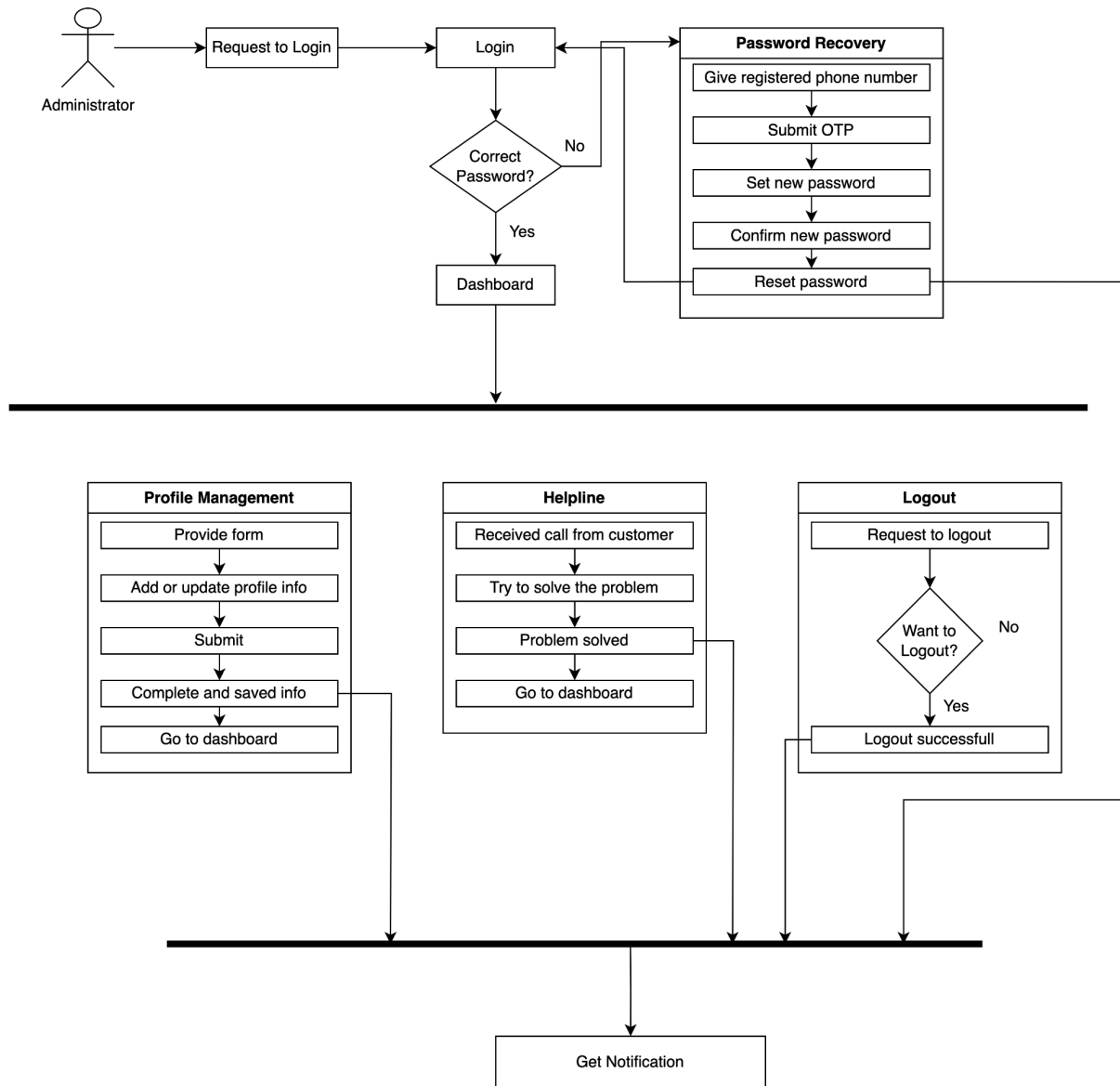


Figure: Block diagram for Administrator

Software Specification Requirement

SRS

FR01	Registration
Description	Users must register to access the system
Stakeholder	Manager, Customer

FR02	Login
Description	Before using users must login their accounts
Stakeholder	Administrator, Manager, Customer

FR03	Profile Management
Description	Users can add, update and manage their profile
Stakeholder	Manager, Customer , Administrator

FR04	Room Booking
Description	Customer can book rooms based on their choice and availability
Stakeholder	Customer

FR05	Room Check-in
Description	Customers will check in when they start their stay.
Stakeholder	Customer

FR06	Room Checkout
Description	Once their stay is over, customers can check out
Stakeholder	Customer

FR07	Generate Bill
Description	Hotel management generate bills based on services used
Stakeholder	Manager

FR08	Pay Bill
Description	Customer can pay their bills
Stakeholder	Customer

FR09	Helpline
Description	The system provides a helpline feature for Customers
Stakeholder	Customer, Manager,Administrator

FR10	Password Recovery
Description	Users can create, update and manage their password
Stakeholder	Manager, Customer, Administrator

FR11	Logout
Description	User can log out of the system
Stakeholder	Administrator, Manager, Customer

Use Case Diagram

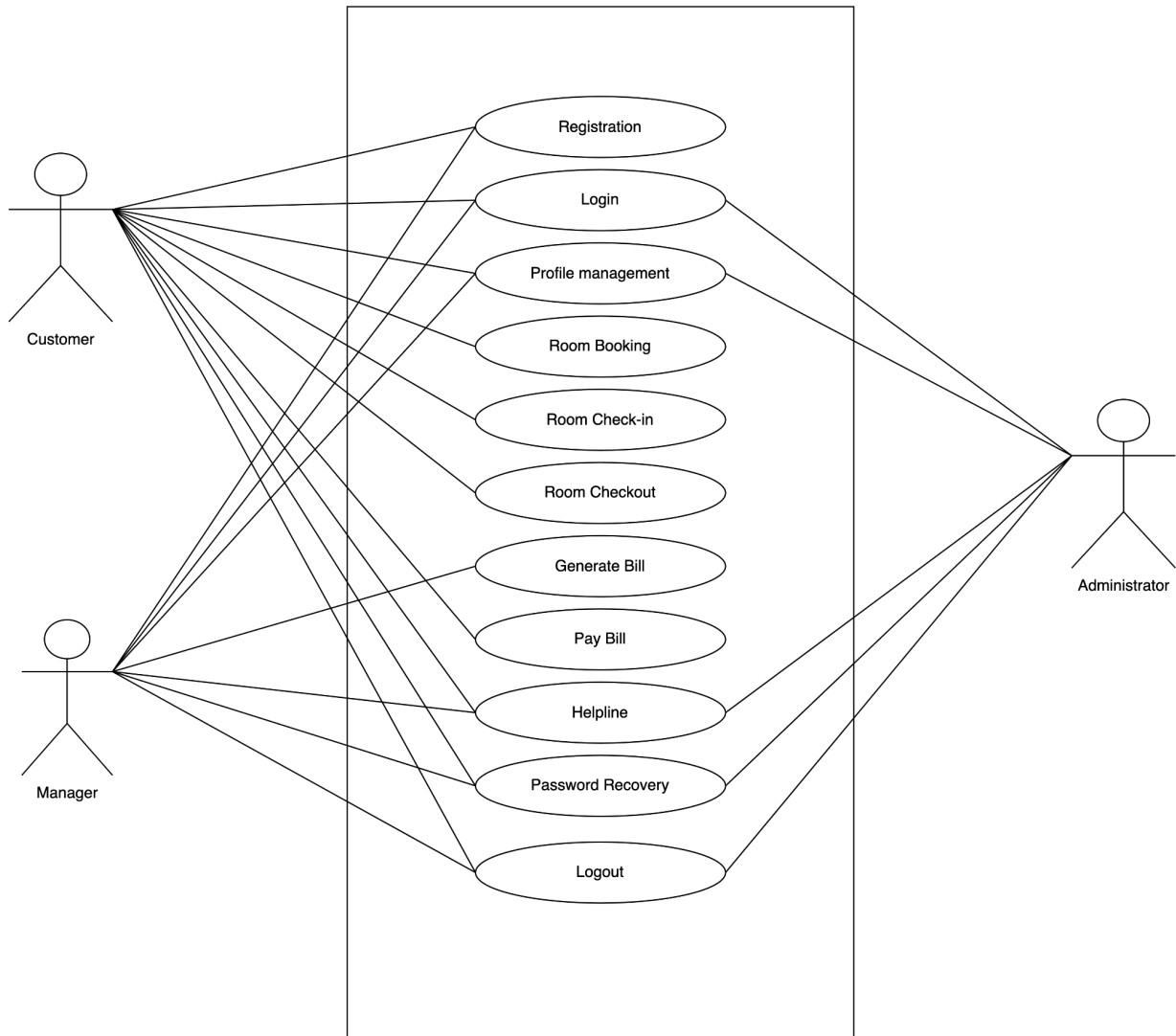


Figure: Use Case Diagram for Hotel Reservation System (Otrip)

User Case Description

Case Description-01 : Registration

Use Case	Registration	
Goal	Users can register to sign in to the Otrip system.	
Precondition	Users must install the Hotel Management app(Otrip) for registration	
Success End Condition	Notification: !Registration Successful!	
Failed End Condition	Notification: !Submission Error!	
Primary Actors: Secondary Actors:	Customer, Manager	
Trigger	User will request a registration form to fill up	
Description / Main Success Scenario	1.	Press “Registration” Button
	2.	Provide registration form
	3.	Enter Information
	4.	Press “Submit” Button.
	5.	Information saved
	6.	Notification: “Registration Successful”
	Alternative Flows	1.1
		1.1.a. Try Again!!
2.1		System Doesn't work
		2.1.a. Try Again Later
3.1		The user did not fill up the details!
		3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.

	4.1	The system did not respond
		4.1.a. Show Error Message.
	5.1	The system doesn't save the details.
		5.1.a. Notification: "Details did not Save"
Quality Requirements	The user will fill up all the details in 30 minutes.	

Case Description-02 : Login

Use Case	Login													
Goal	Users will login to enter the Otrip system.													
Precondition	Users must be registered into Otrip													
Success End Condition	Notification: !Login Successful!													
Failed End Condition	Notification:!!Login Failed!													
Primary Actors: Secondary Actors:	Customer, Manager System Administration													
Trigger	User will request to log into the app													
Description / Main Success Scenario	<table><tr><td>1.</td><td>Press “Login” Button</td></tr><tr><td>2.</td><td>Provide login page</td></tr><tr><td>3.</td><td>Enter Information(ID and Password)</td></tr><tr><td>4.</td><td>Press “Submit” Button.</td></tr><tr><td>5.</td><td>Information Verified and login</td></tr><tr><td>6.</td><td>Notification: “Login Successful”</td></tr></table>		1.	Press “Login” Button	2.	Provide login page	3.	Enter Information(ID and Password)	4.	Press “Submit” Button.	5.	Information Verified and login	6.	Notification: “Login Successful”
1.	Press “Login” Button													
2.	Provide login page													
3.	Enter Information(ID and Password)													
4.	Press “Submit” Button.													
5.	Information Verified and login													
6.	Notification: “Login Successful”													
Alternative Flows	<table><tr><td>1.1</td><td>System Error</td></tr></table>		1.1	System Error										
1.1	System Error													

		1.1.a. Try Again!!
	2.1	System Doesn't work
		2.1.a. Try Again Later
	3.1	The user did not fill in the ID and password!
		3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.
	4.1	The system did not respond
		4.1.a. Show Error Message.
	5.1	The password is wrong.
		5.1.a. Notification: “Wrong Password.Try again”
	5.2	The ID is wrong.
		5.2.a. Notification: “The ID is incorrect.Try again”
Quality Requirements	The user will fill up login info in 20 minutes	

Case Description-03 : Profile Management

Use Case	Profile Management			
Goal	Users will give required information to manage their account in the Otrip system.			
Precondition	Users must log in to the system			
Success End Condition	Notification: !Profile Updated!			
Failed End Condition	Notification: !Request Failed!			
Primary Actors: Secondary Actors:	Customer, Manager System Administration			
Trigger	User will request Profile Creation form to fill up			
Description / Main Success Scenario	<table><tr><td>1.</td><td>Press “Create Profile” Button</td></tr></table>		1.	Press “Create Profile” Button
1.	Press “Create Profile” Button			

	<table> <tr> <td>2.</td><td>Provide profile form</td></tr> <tr> <td>3.</td><td>Enter Information</td></tr> <tr> <td>4.</td><td>Press “Submit” Button.</td></tr> <tr> <td>5.</td><td>Information saved</td></tr> <tr> <td>6.</td><td>Notification: “Profile Creation Successful”</td></tr> </table>	2.	Provide profile form	3.	Enter Information	4.	Press “Submit” Button.	5.	Information saved	6.	Notification: “Profile Creation Successful”										
2.	Provide profile form																				
3.	Enter Information																				
4.	Press “Submit” Button.																				
5.	Information saved																				
6.	Notification: “Profile Creation Successful”																				
Alternative Flows	<table> <tr> <td>1.1</td><td>System Error</td></tr> <tr> <td></td><td>1.1.a. Try Again!!</td></tr> <tr> <td>2.1</td><td>System Doesn't work</td></tr> <tr> <td></td><td>2.1.a. Try Again Later</td></tr> <tr> <td>3.1</td><td>The user did not fill up the details!</td></tr> <tr> <td></td><td>3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.</td></tr> <tr> <td>4.1</td><td>The system did not respond</td></tr> <tr> <td></td><td>4.1.a. Show Error Message.</td></tr> <tr> <td>5.1</td><td>The system doesn't save the details.</td></tr> <tr> <td></td><td>5.1.a. Notification: “Details did not Save”</td></tr> </table>	1.1	System Error		1.1.a. Try Again!!	2.1	System Doesn't work		2.1.a. Try Again Later	3.1	The user did not fill up the details!		3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.	4.1	The system did not respond		4.1.a. Show Error Message.	5.1	The system doesn't save the details.		5.1.a. Notification: “Details did not Save”
1.1	System Error																				
	1.1.a. Try Again!!																				
2.1	System Doesn't work																				
	2.1.a. Try Again Later																				
3.1	The user did not fill up the details!																				
	3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.																				
4.1	The system did not respond																				
	4.1.a. Show Error Message.																				
5.1	The system doesn't save the details.																				
	5.1.a. Notification: “Details did not Save”																				
Quality Requirements	The user Will fill up all the details in 30 minutes.																				

Case Description-04 : Room Booking

Use Case	Room Booking
Goal	Users will book room according to their necessity
Precondition	Users must log in to the system
Success End Condition	Notification: !Booking Successful!
Failed End	Notification: !Request Failed!

Condition																													
Primary Actors: Secondary Actors:	Customer																												
Trigger	User will request to book a room																												
Description / Main Success Scenario	<table> <tr><td>1.</td><td>Press “Book” Button</td></tr> <tr><td>2.</td><td>Provide Room Booking form</td></tr> <tr><td>3.</td><td>Enter Information</td></tr> <tr><td>4.</td><td>Press “Submit” Button.</td></tr> <tr><td>5.</td><td>Information saved</td></tr> <tr><td>6.</td><td>Notification: !Booking Successful!</td></tr> </table>	1.	Press “Book” Button	2.	Provide Room Booking form	3.	Enter Information	4.	Press “Submit” Button.	5.	Information saved	6.	Notification: !Booking Successful!																
1.	Press “Book” Button																												
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Alternative Flows	<table> <tr><td>1.1</td><td>System Error</td></tr> <tr><td></td><td>1.1.a. Try Again!!</td></tr> <tr><td>2.1</td><td>System Doesn't work</td></tr> <tr><td></td><td>2.1.a. Try Again Later</td></tr> <tr><td>3.1</td><td>The user did not fill up the details!</td></tr> <tr><td></td><td>3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.</td></tr> <tr><td>4.1</td><td>The system did not respond</td></tr> <tr><td></td><td>4.1.a. Show Error Message.</td></tr> <tr><td>5.1</td><td>The system doesn't save the details.</td></tr> <tr><td></td><td>5.1.a. Notification: “Details did not Save. Please try again”</td></tr> <tr><td>5.2</td><td>The user gave an unavailable room number.</td></tr> <tr><td></td><td>5.2.a. Notification: !The room is unavailable!</td></tr> <tr><td>5.3</td><td>The user gave an invalid room number.</td></tr> <tr><td></td><td>5.3.a. Notification: !Invalid Room number!</td></tr> </table>	1.1	System Error		1.1.a. Try Again!!	2.1	System Doesn't work		2.1.a. Try Again Later	3.1	The user did not fill up the details!		3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.	4.1	The system did not respond		4.1.a. Show Error Message.	5.1	The system doesn't save the details.		5.1.a. Notification: “Details did not Save. Please try again”	5.2	The user gave an unavailable room number.		5.2.a. Notification: !The room is unavailable!	5.3	The user gave an invalid room number.		5.3.a. Notification: !Invalid Room number!
1.1	System Error																												
	1.1.a. Try Again!!																												
2.1	System Doesn't work																												
	2.1.a. Try Again Later																												
3.1	The user did not fill up the details!																												
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4.1	The system did not respond																												
	4.1.a. Show Error Message.																												
5.1	The system doesn't save the details.																												
	5.1.a. Notification: “Details did not Save. Please try again”																												
5.2	The user gave an unavailable room number.																												
	5.2.a. Notification: !The room is unavailable!																												
5.3	The user gave an invalid room number.																												
	5.3.a. Notification: !Invalid Room number!																												

Quality Requirements	The user Will fill up all the required information in 30 minutes.

Case Description-05 : Room Check-in

Use Case	Room Check-in	
Goal	Users will check-in the Otrip system on arrival	
Precondition	The room must be booked	
Success End Condition	Notification: !Check-in Successful!	
Failed End Condition	Notification: !Action Failed!	
Primary Actors: Secondary Actors:	Customer	
Trigger	User will request a Check-in for their booked room	
Description / Main Success Scenario	1.	Press “Check-in” Button
	2.	Provide check-in form
	3.	Enter Information
	4.	Press “Submit” Button.
	5.	Information verified and saved
	6.	Notification: !Check-in Successful!
	Alternative Flows	1.1
		1.1.a. Try Again!!
2.1		System Doesn't work
		2.1.a. Try Again Later
3.1		The user did not fill up the details!

		3.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.
	4.1	The system did not respond
		4.1.a. Show Error Message.
	5.1	The system doesn't save the details.
		5.1.a. Notification: “Details did not Save”
Quality Requirements	The user will confirm check-in in 10 minutes after check-in request	

Case Description-06 : Room Check-out

Use Case	Room Checkout	
Goal	Users(Customer) will leave the room and check-out using Otrip	
Precondition	The room must be booked.	
Success End Condition	Notification: !Checking-out Successful!	
Failed End Condition	Notification: !Checking-out Failed!	
Primary Actors: Secondary Actors:	Customer	
Trigger	User will request a room check-out	
Description / Main Success Scenario	1.	Press “Check-out” Button
	2.	Provide check-out form
	3.	Enter required Information
	4.	Press “Submit” Button.
	5.	Information Verified
	6.	Notification: !Check-out Completed!
	Alternative Flows	

	1.1	System Error
		1.1.a. Try Again!!
	2.1	System Doesn't work
		2.1.a. Try Again Later
	3.1	The user did not give required information!
		3.1.a. Checked By the system & Notify by "Please! Input valid room information".
	5.1	The user did not give valid information!
		5.1.a. Notification: "Please! Input valid Information"
	6.1	The system does not complete the check-out .
		6.1.a. Notification: "Check-out Failed"
Quality Requirements	The user will put a customer id in 30 minutes.	

Case Description-07: Generate Bill

Use Case	Generate Bill	
Goal	Users(Manager) can generate customer bill by using Otrip	
Precondition	Users must be logged into the system	
Success End Condition	Notification: !Bill Generated!	
Failed End Condition	Notification: !Bill Generation Failed.Try Again!	
Primary Actors: Secondary Actors:	Manager	
Trigger	User will request a Bill Generation	
Description / Main Success Scenario		
	1.	Press “Generate Bill” Button
	2.	Provide Customer ID

	<table> <tr> <td>3.</td><td>Enter room Information</td></tr> <tr> <td>4.</td><td>Press "Submit" Button.</td></tr> <tr> <td>5.</td><td>Bill generation processing</td></tr> <tr> <td>6.</td><td>The system generates the bill</td></tr> </table>	3.	Enter room Information	4.	Press "Submit" Button.	5.	Bill generation processing	6.	The system generates the bill																
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4.	Press "Submit" Button.																								
5.	Bill generation processing																								
6.	The system generates the bill																								
Alternative Flows	<table> <tr> <td>1.1</td><td>System Error</td></tr> <tr> <td></td><td>1.1.a. Try Again!!</td></tr> <tr> <td>2.1</td><td>System Doesn't work</td></tr> <tr> <td></td><td>2.1.a. Try Again Later</td></tr> <tr> <td>3.1</td><td>The user did not give room information!</td></tr> <tr> <td></td><td>3.1.a. Checked By the system & Notify by "Please! Input valid room information".</td></tr> <tr> <td>3.2</td><td>The user did not give Customer ID!</td></tr> <tr> <td></td><td>3.2.a. Checked By the system & Notify by "Please! Input valid Customer ID"</td></tr> <tr> <td>4.1</td><td>The system did not respond</td></tr> <tr> <td></td><td>4.1.a. Show Error Message.</td></tr> <tr> <td>6.1</td><td>The system Doesn't Generate the bill.</td></tr> <tr> <td></td><td>6.1.a. Notification: "Bill Generation Failed"</td></tr> </table>	1.1	System Error		1.1.a. Try Again!!	2.1	System Doesn't work		2.1.a. Try Again Later	3.1	The user did not give room information!		3.1.a. Checked By the system & Notify by "Please! Input valid room information".	3.2	The user did not give Customer ID!		3.2.a. Checked By the system & Notify by "Please! Input valid Customer ID"	4.1	The system did not respond		4.1.a. Show Error Message.	6.1	The system Doesn't Generate the bill.		6.1.a. Notification: "Bill Generation Failed"
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	4.1.a. Show Error Message.																								
6.1	The system Doesn't Generate the bill.																								
	6.1.a. Notification: "Bill Generation Failed"																								
Quality Requirements	The user will put a Customer id in 30 minutes.																								

Case Description-08 : Pay Bill

Use Case	Pay Bill
Goal	Users(Customer) will pay the bill using Otrip
Precondition	Must be logged in the app The room must be booked

Success End Condition	Notification: !Payment Successful!												
Failed End Condition	Notification: !Payment Failed!												
Primary Actors: Secondary Actors:	Customer												
Trigger	User will request a bill payment												
Description / Main Success Scenario	<table> <tr> <td>1.</td><td>Press "Pay Bill" Button</td></tr> <tr> <td>2.</td><td>Select Payment Option</td></tr> <tr> <td>3.</td><td>Pay the bill</td></tr> <tr> <td>4.</td><td>Press "Submit" Button.</td></tr> <tr> <td>5.</td><td>Notification: !Payment Successful!</td></tr> </table>	1.	Press "Pay Bill" Button	2.	Select Payment Option	3.	Pay the bill	4.	Press "Submit" Button.	5.	Notification: !Payment Successful!		
1.	Press "Pay Bill" Button												
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Alternative Flows	<table> <tr> <td>1.1</td><td>System Error</td></tr> <tr> <td></td><td>1.1.a. Try Again!!</td></tr> <tr> <td>2.1</td><td>The user did not choose a payment option.</td></tr> <tr> <td></td><td>2.1.a. Notification: !Please choose a payment option!</td></tr> <tr> <td>4.1</td><td>The system did not respond</td></tr> <tr> <td></td><td>4.1.a. Show Error Message.</td></tr> </table>	1.1	System Error		1.1.a. Try Again!!	2.1	The user did not choose a payment option.		2.1.a. Notification: !Please choose a payment option!	4.1	The system did not respond		4.1.a. Show Error Message.
1.1	System Error												
	1.1.a. Try Again!!												
2.1	The user did not choose a payment option.												
	2.1.a. Notification: !Please choose a payment option!												
4.1	The system did not respond												
	4.1.a. Show Error Message.												
Quality Requirements	The user will put account information in 30 minutes.												

Case Description-09 : Helpline

Use Case	Helpline
Goal	Provide Customers with access to a helpline service for assistance.
Precondition	Customers must be logged into the system or have access to the helpline.

Success End Condition	Customers receive assistance or are connected to the right department.	
Failed End Condition	Notification: !Try Again!	
Primary Actors Secondary Actors	Customer, Manager Administrator	
Trigger	Customers request assistance through the helpline.	
Main Success Scenario	01	Customers click the “Helpline” button.
	02	The customer submits a query or selects a topic.
	03	The system connects to the appropriate department.
	04	Customers receive assistance.
Alternative Flows	3.1	The system fails to connect to the department.
		3.1.a. Notification: !Try Again!
Quality Requirements	The system must be available 24/7.	

Case Description-10 : Password Recovery

Use Case	Password Recovery
Goal	Allow users to recover or reset their password.
Precondition	Users must have an account in the system.
Success End Condition	Notification: !Password Reset Successful!
Failed End Condition	Notification: !Try Again!
Primary Actors Secondary Actors	Manager, Customer Administrator

Trigger	The user selects the Forget Password option.	
Main Success Scenario	01	The user selects the “Forgot Password” option.
	02	The user enters email/ phone.
	03	The system sends OTP.
	04	User resets password.
	05	Password is updated.
Alternative Flows	2.1	Email/ phone is unregistered.
		2.1.a.Notification: “Invalid Email/ Phone Number”
Quality Requirements	Passwords must be updated within 10 minutes after the OTP is sent.	

Case Description-11 : Logout

Use Case	Logout	
Goal	Users can securely log out the system.	
Precondition	Users are logged into the system.	
Success End Condition	Notification: !Logout Successful!	
Failed End Condition	Notification: !Logout Failed!	
Primary Actors Secondary Actors	Manager, Customer Administrator	
Trigger	The user selects the “Logout” option.	
Main Success Scenario	01	User selects “Logout”
	02	System confirms logout.
	03	The user is logged out.

Alternative Flows	3.1	The system fails to log out.
		3.1.a. Notification: !Try Again!
Quality Requirements	The system should log out within 5 seconds of the request.	

System Design

Data Flow Diagram

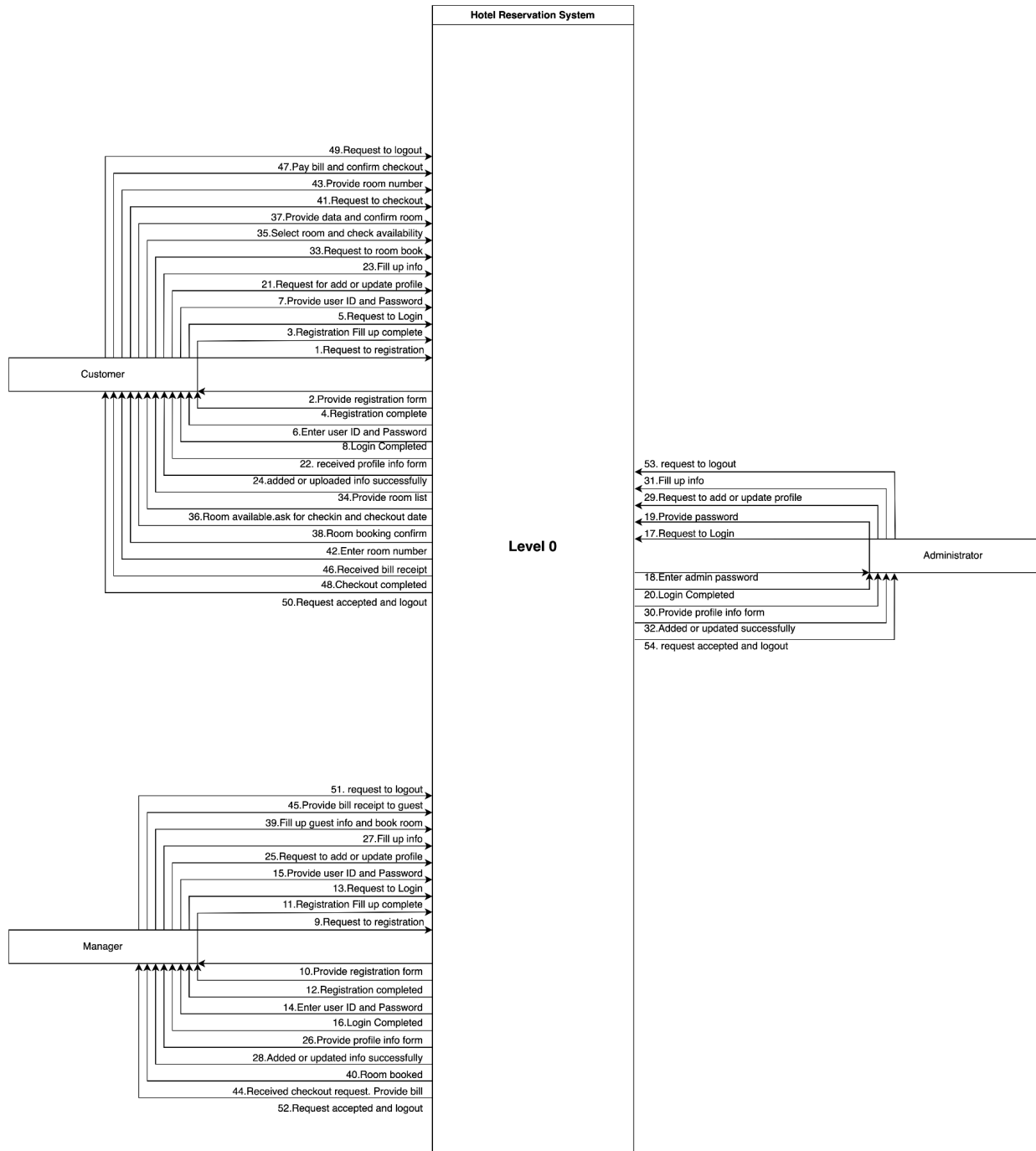


Figure: Data Flow Diagram level 0

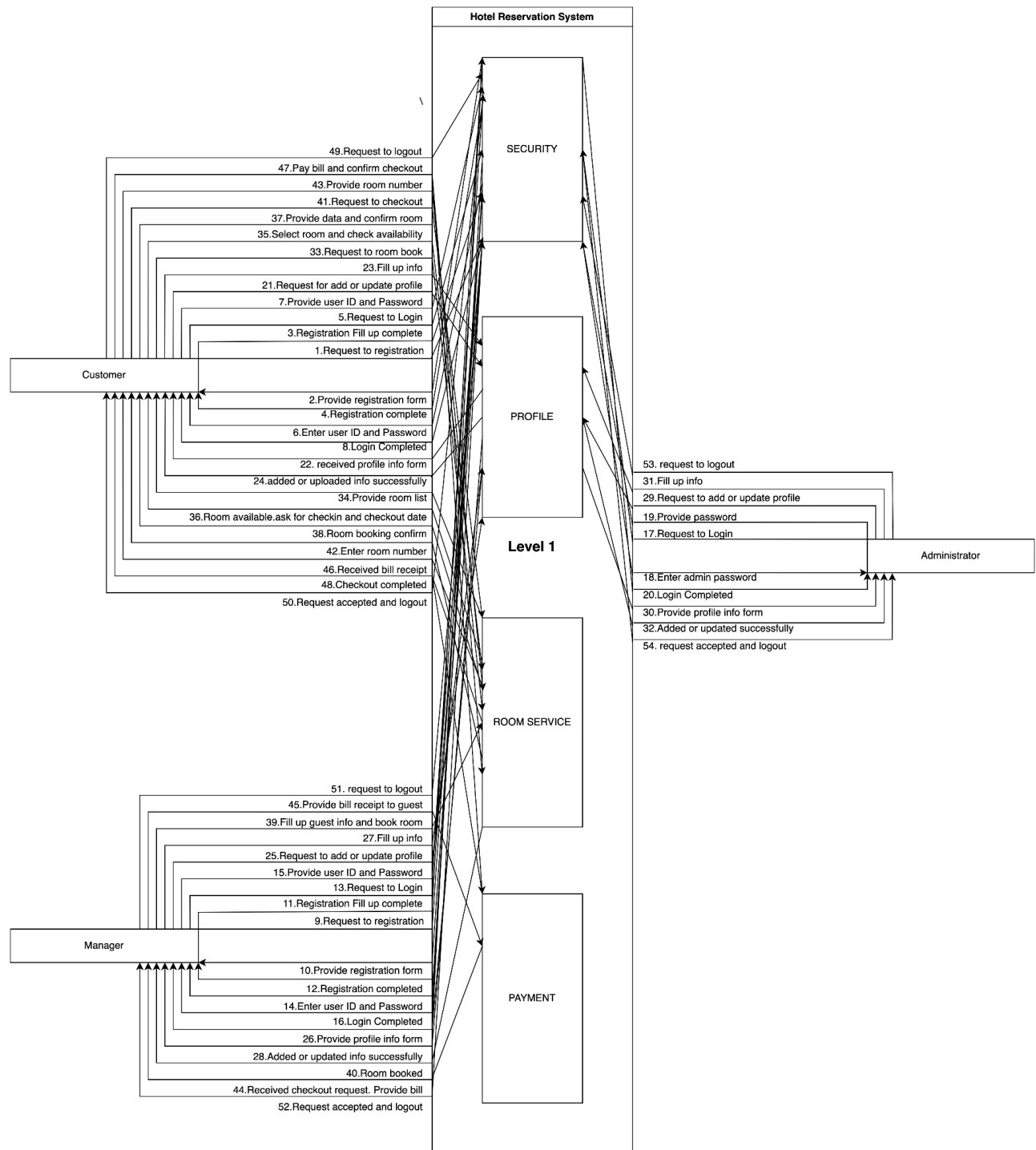
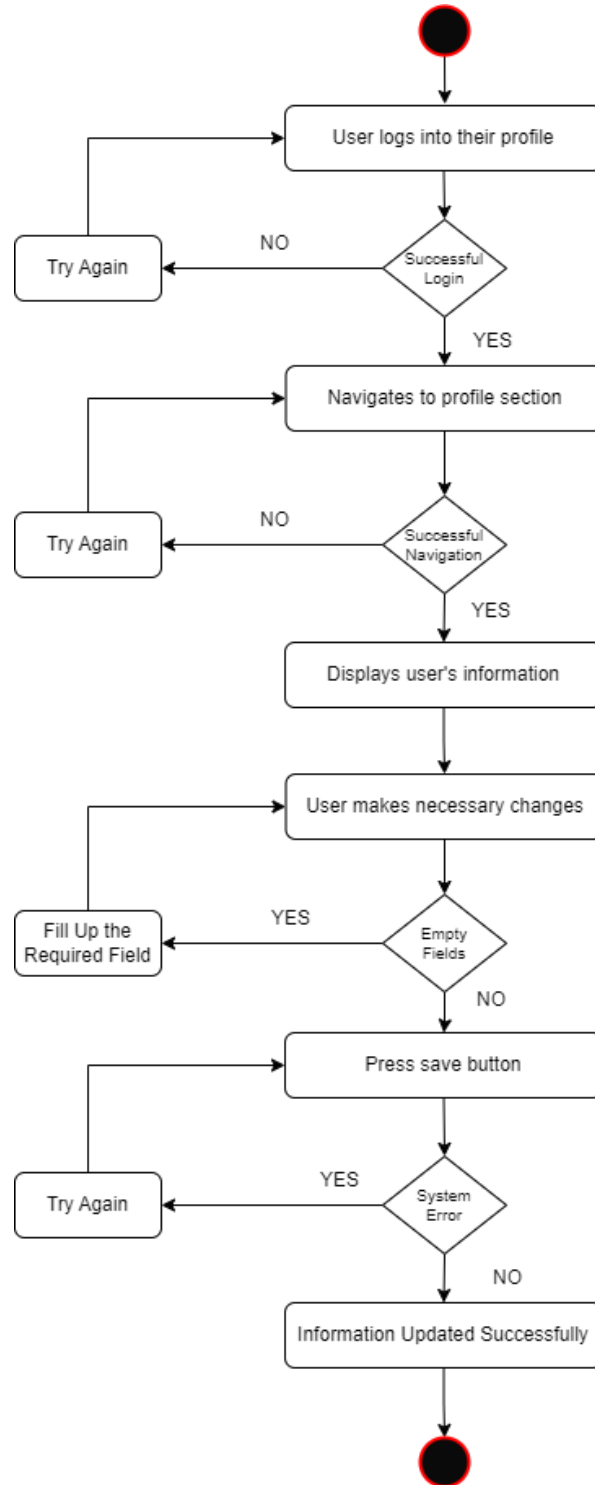


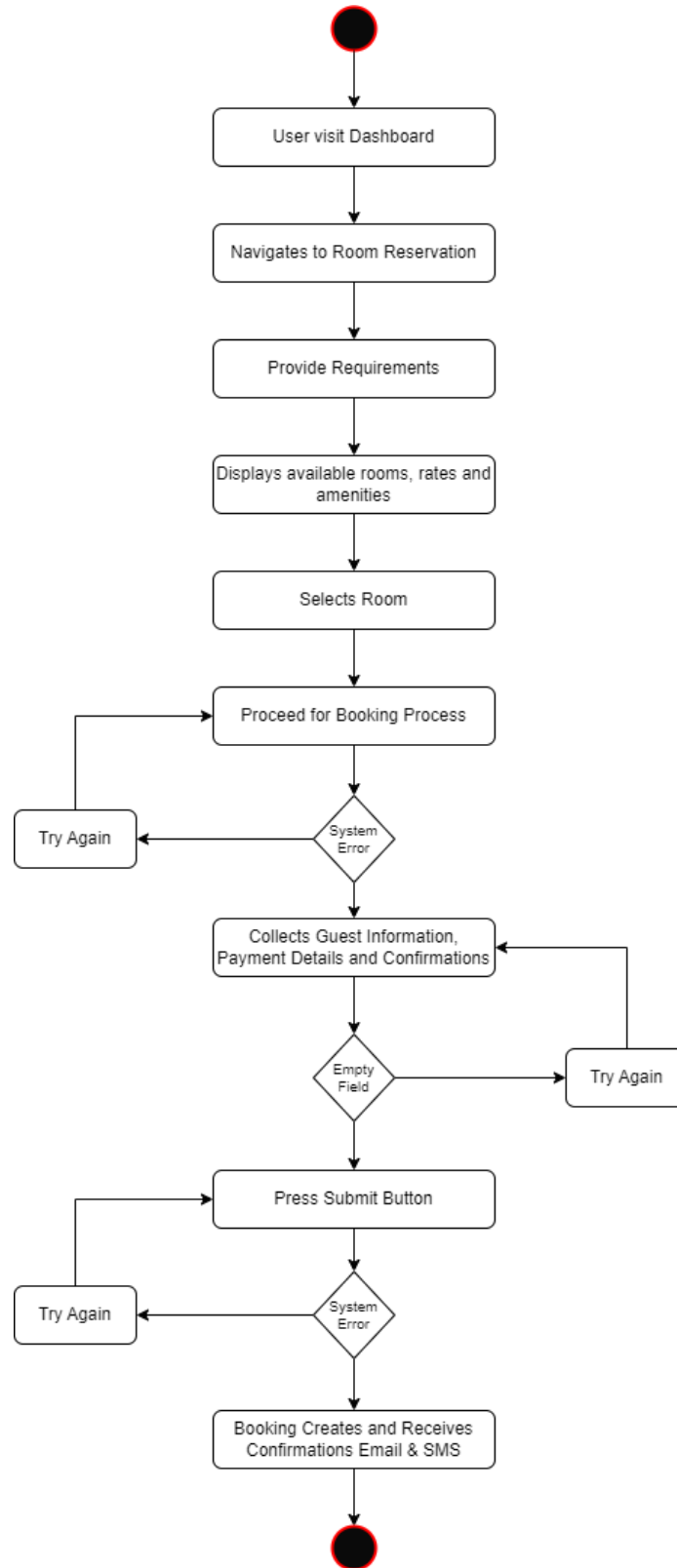
Figure: Data Flow Diagram Level 1

Activity Diagram

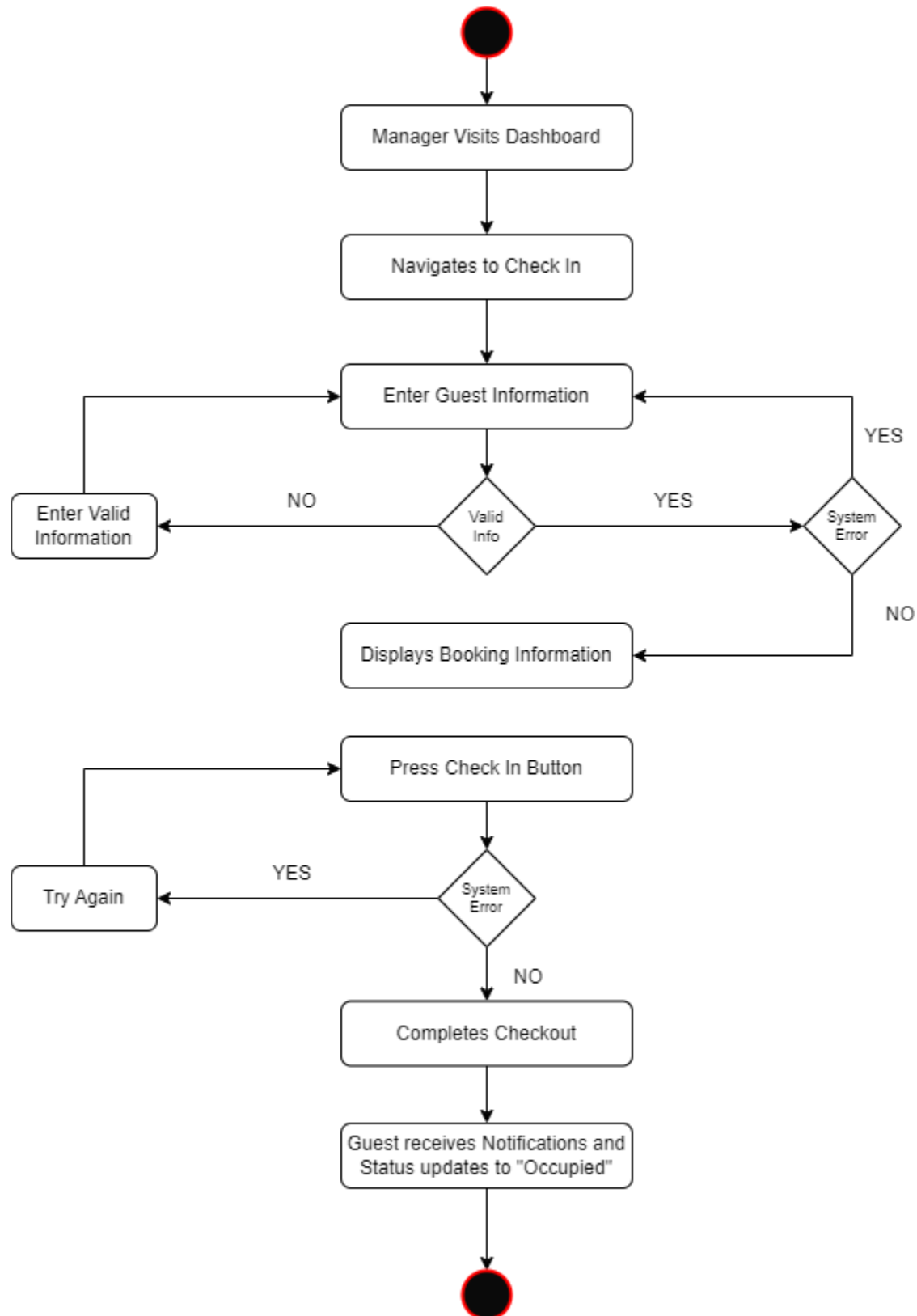
Activity Diagram - 01: Profile Management



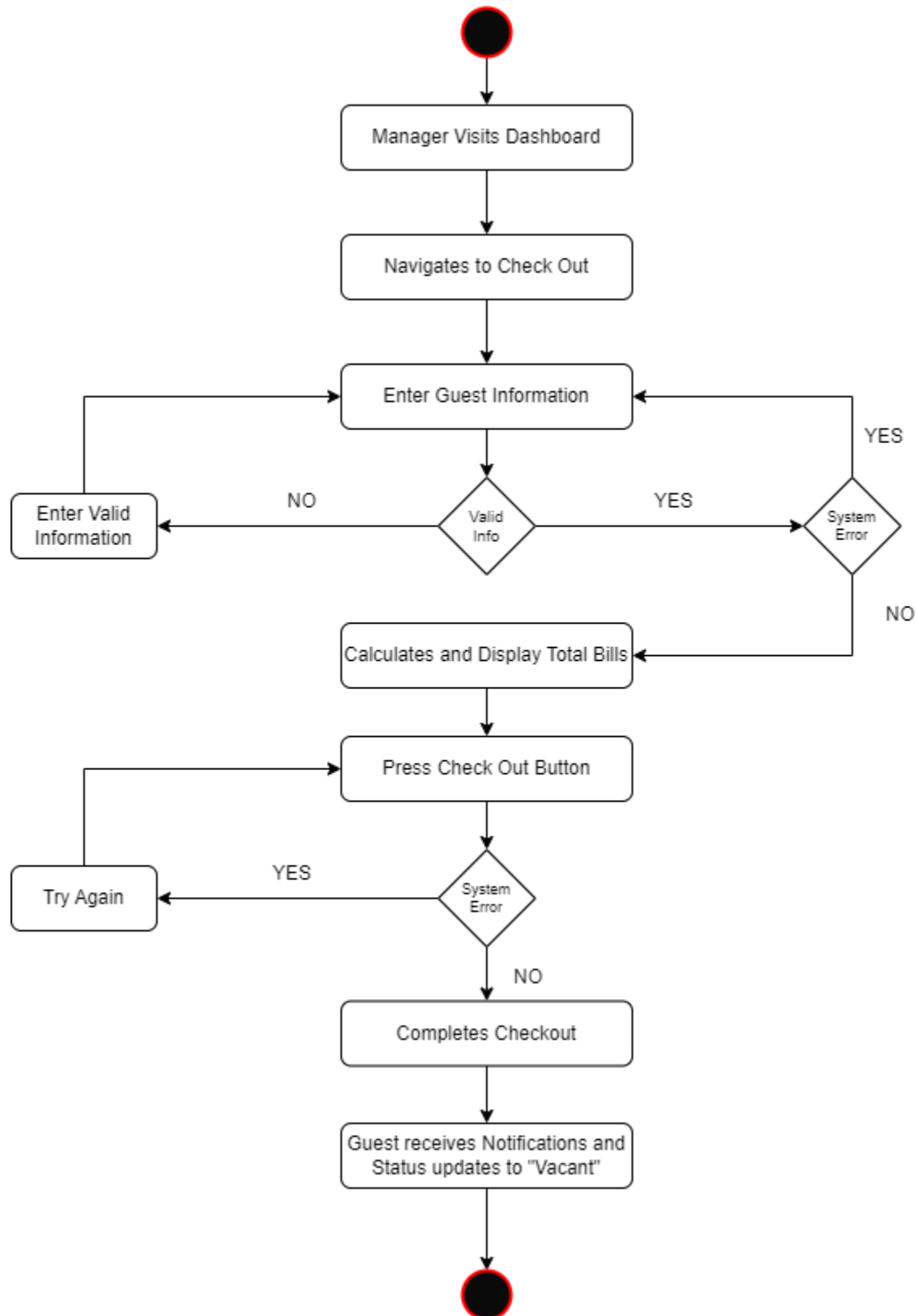
Activity Diagram - 02: Room Reservation



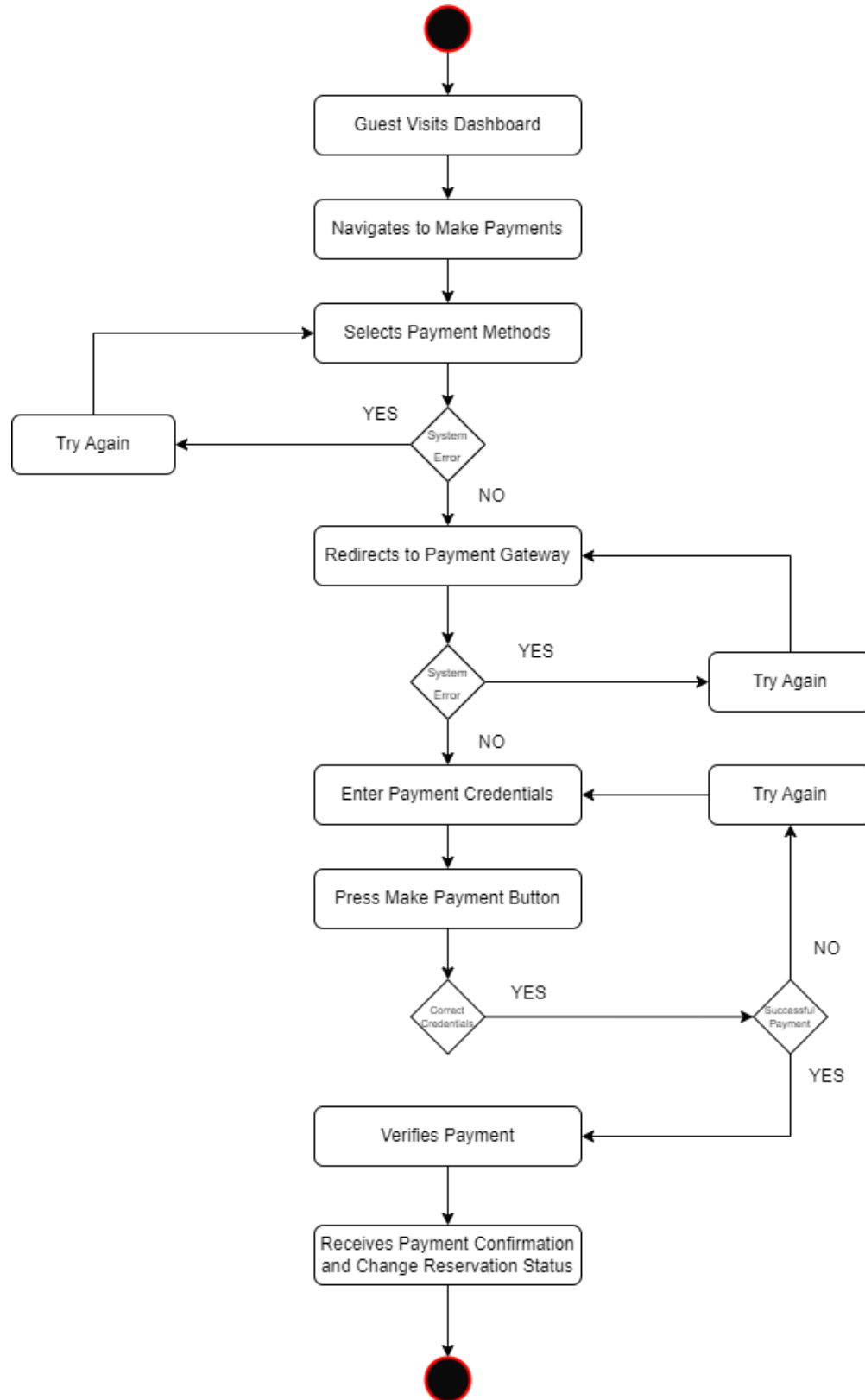
Activity Diagram - 03: Check In



Activity Diagram - 04: Check Out

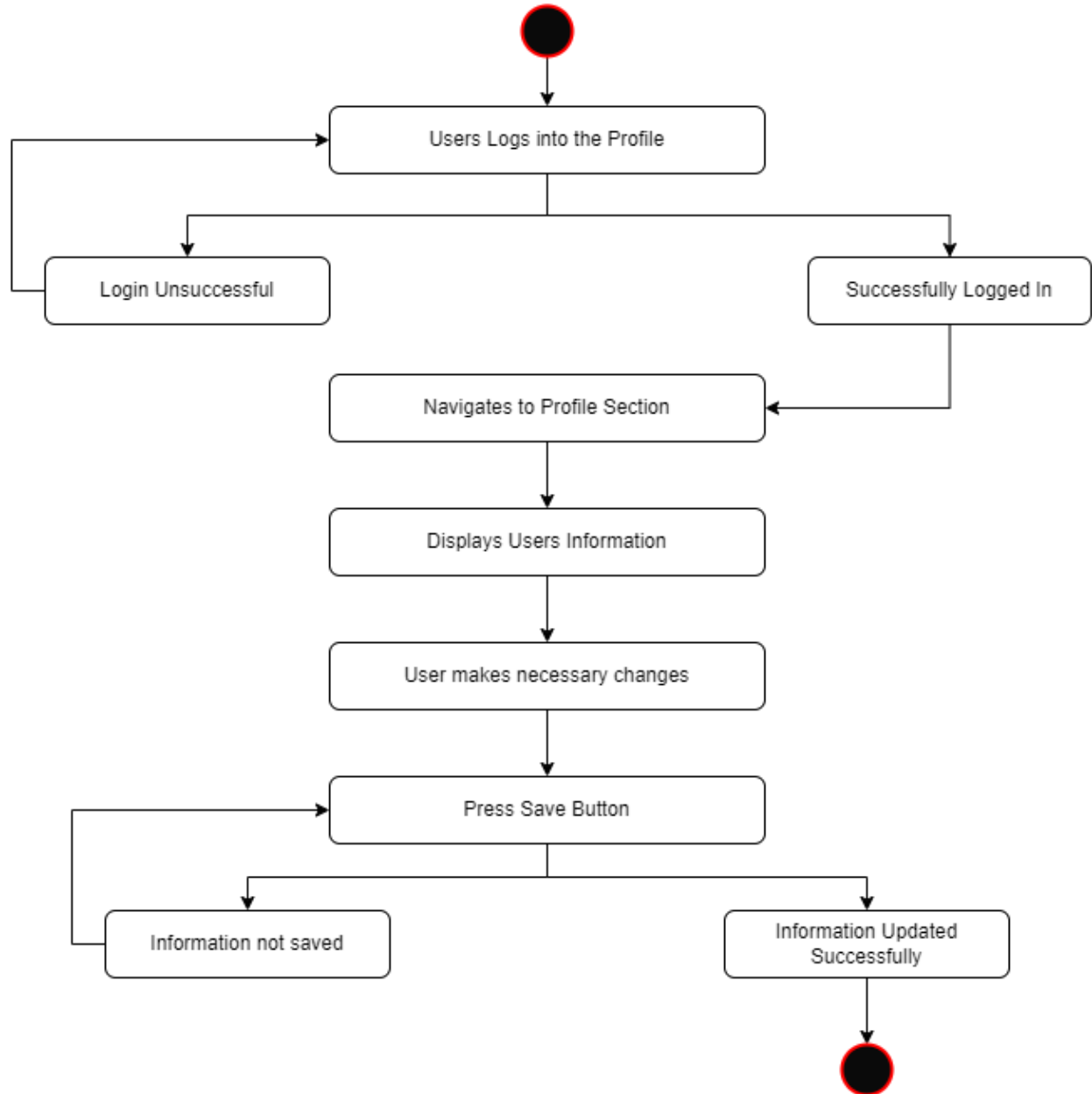


Activity Diagram - 05: Payment System

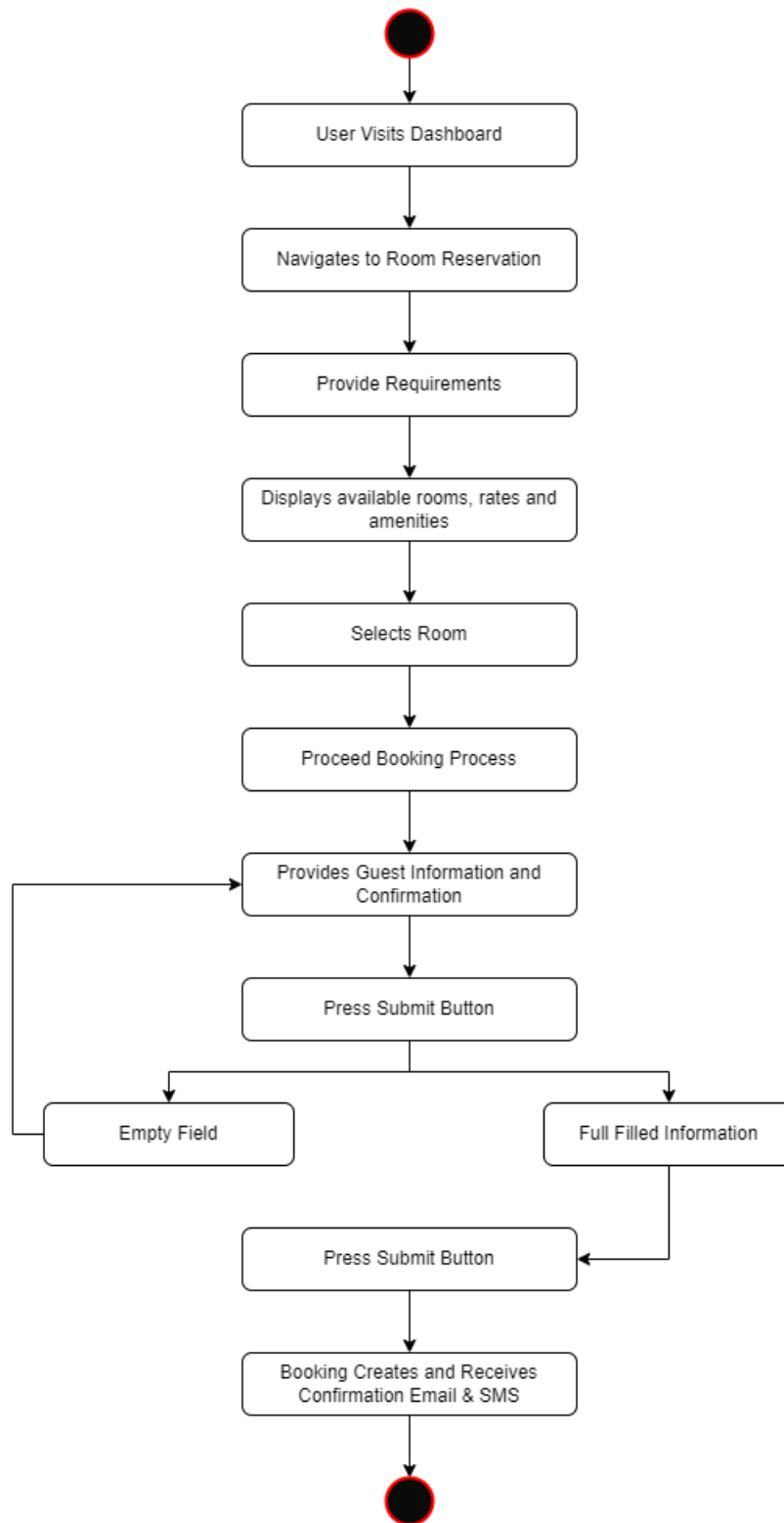


State Diagram

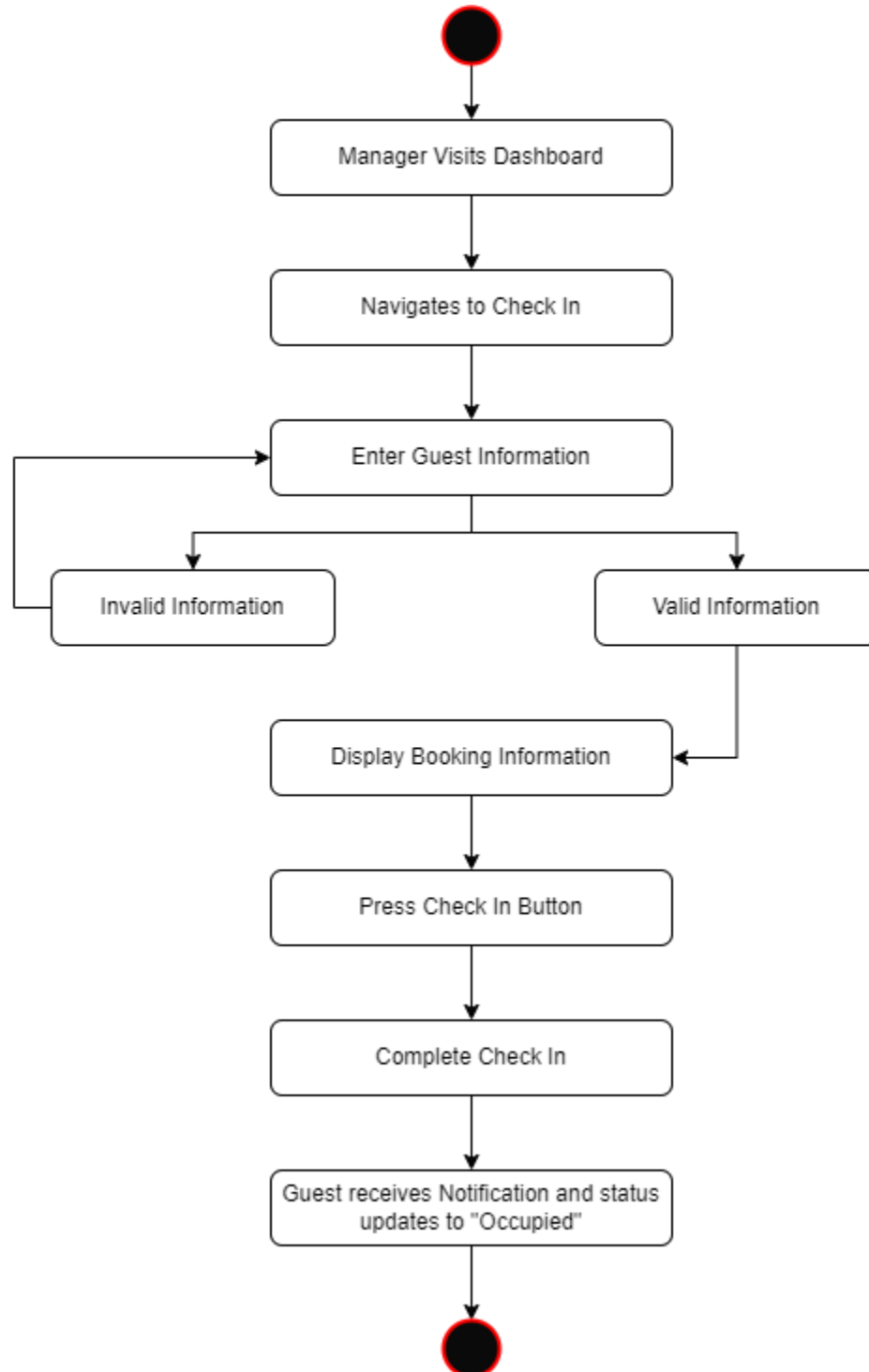
State Diagram - 01: Profile Management



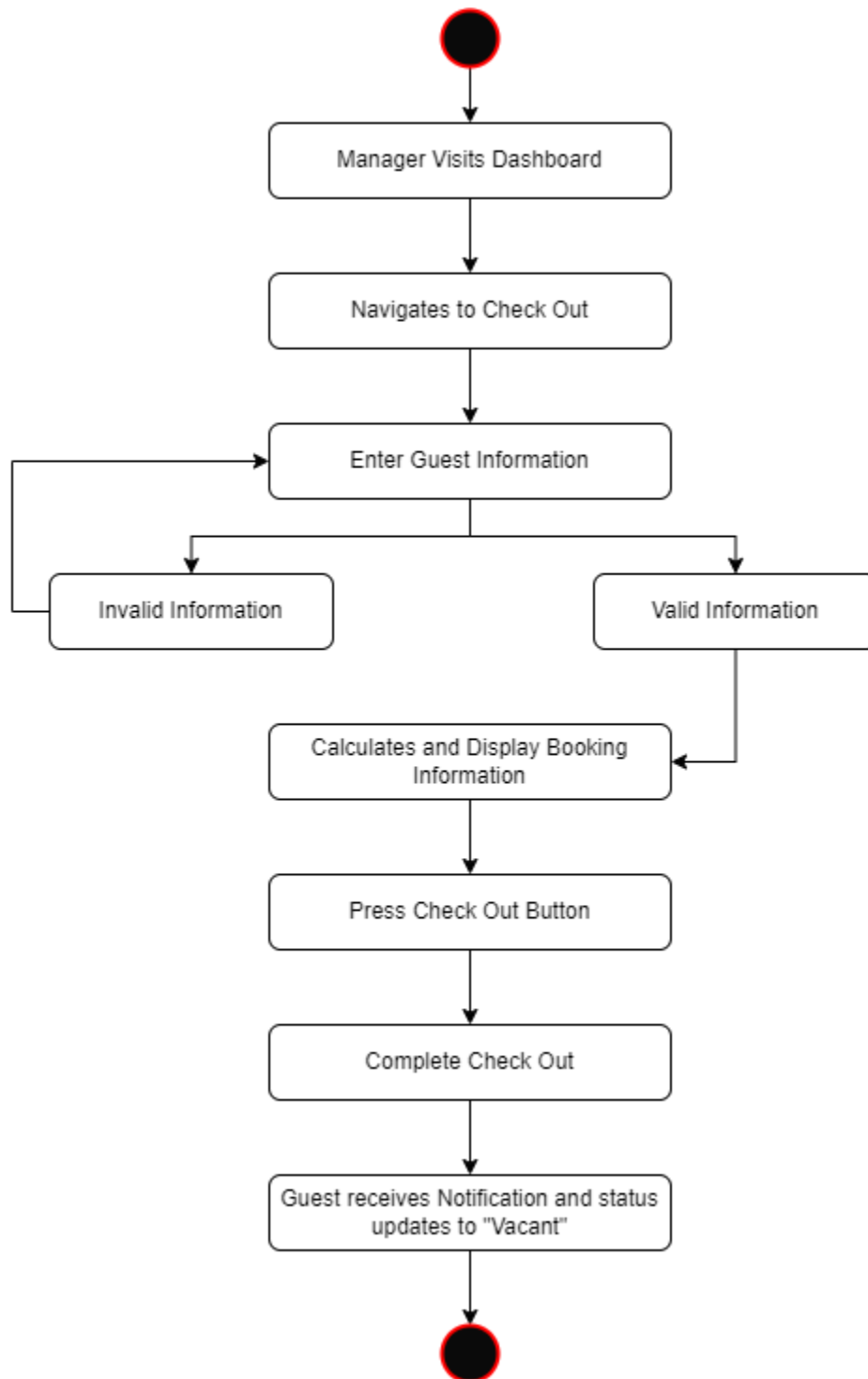
State Diagram - 02: Room Reservation



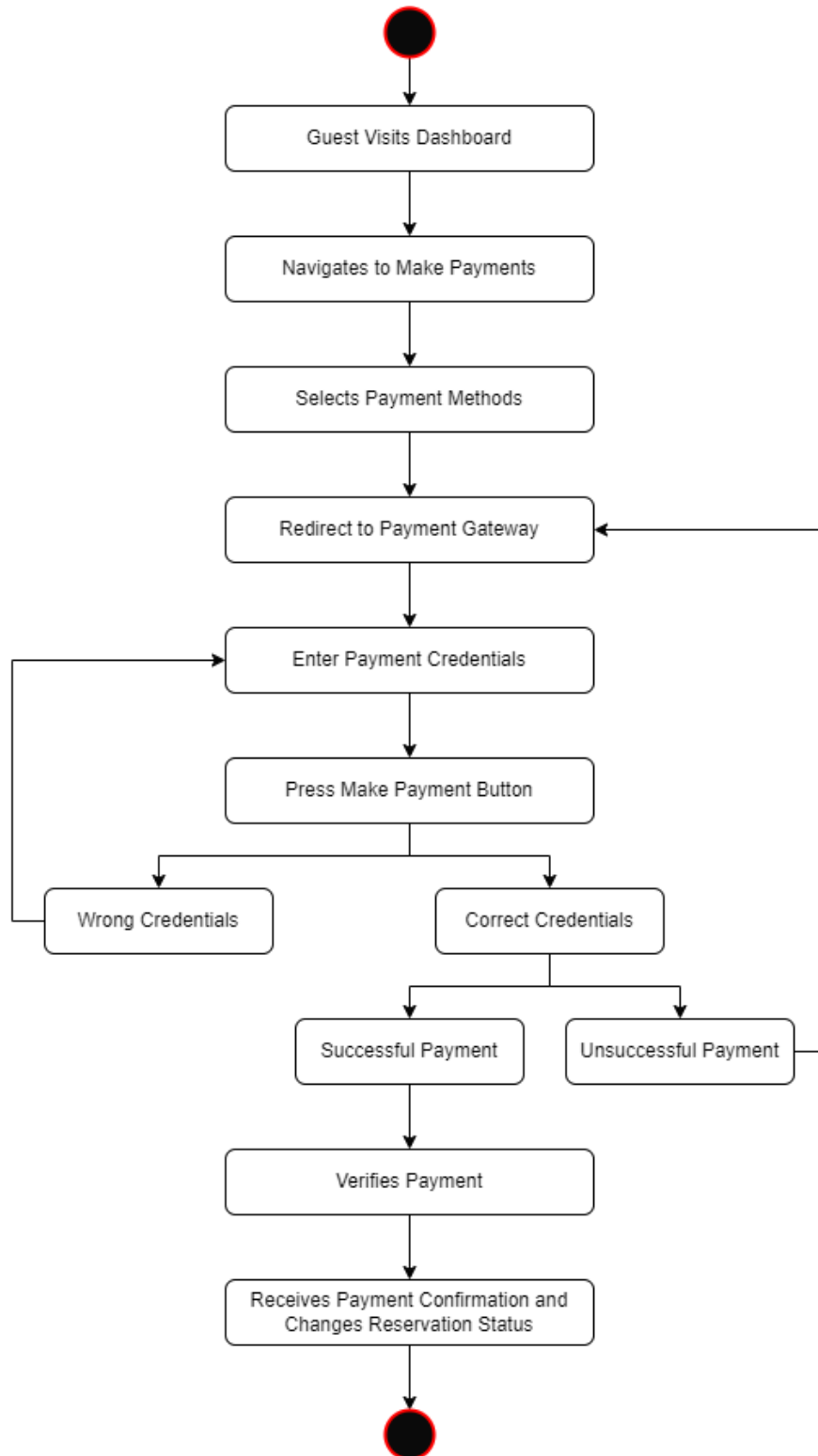
State Diagram - 03: Check In



State Diagram - 04: Check Out

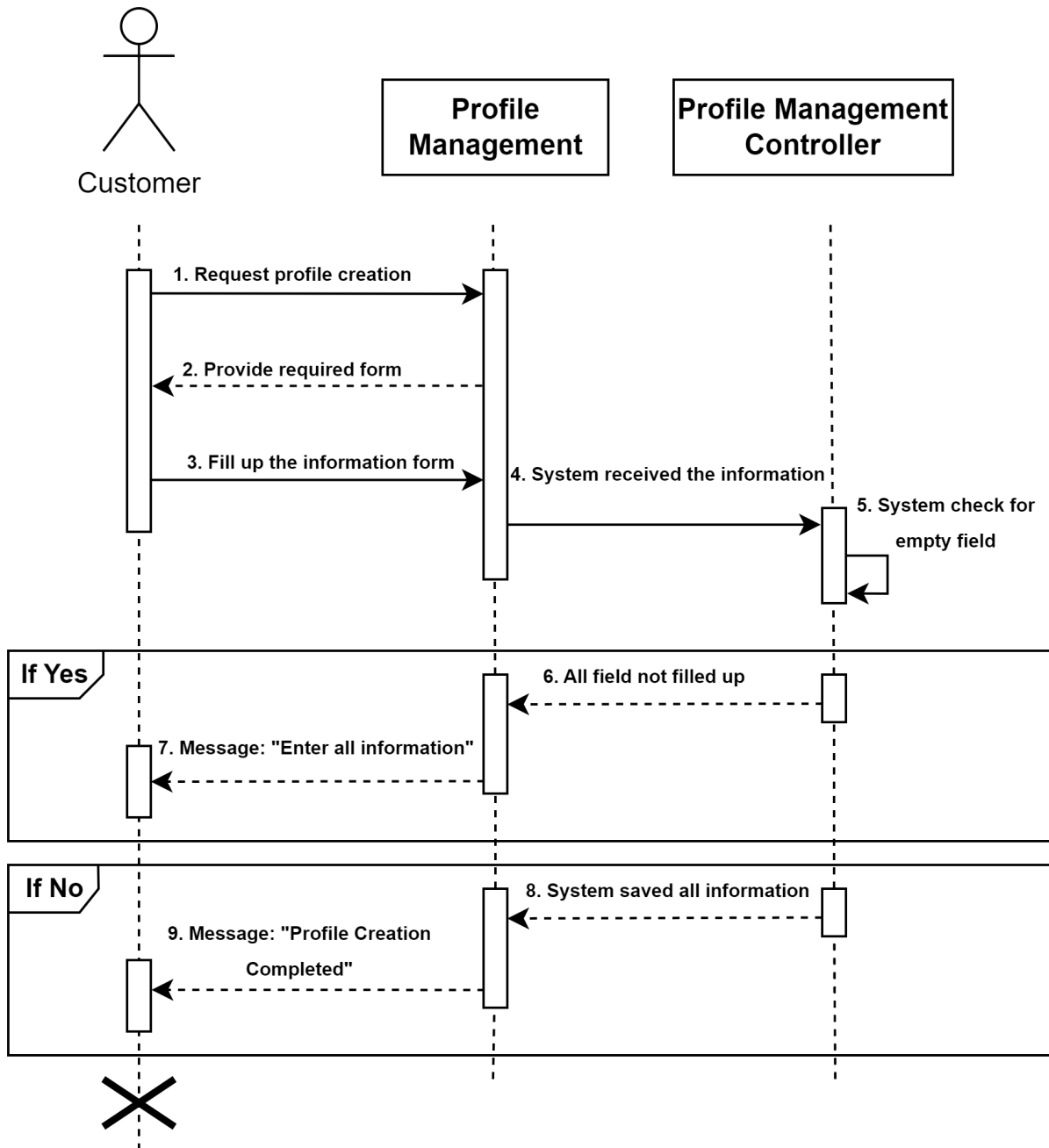


State Diagram - 05: Payment System

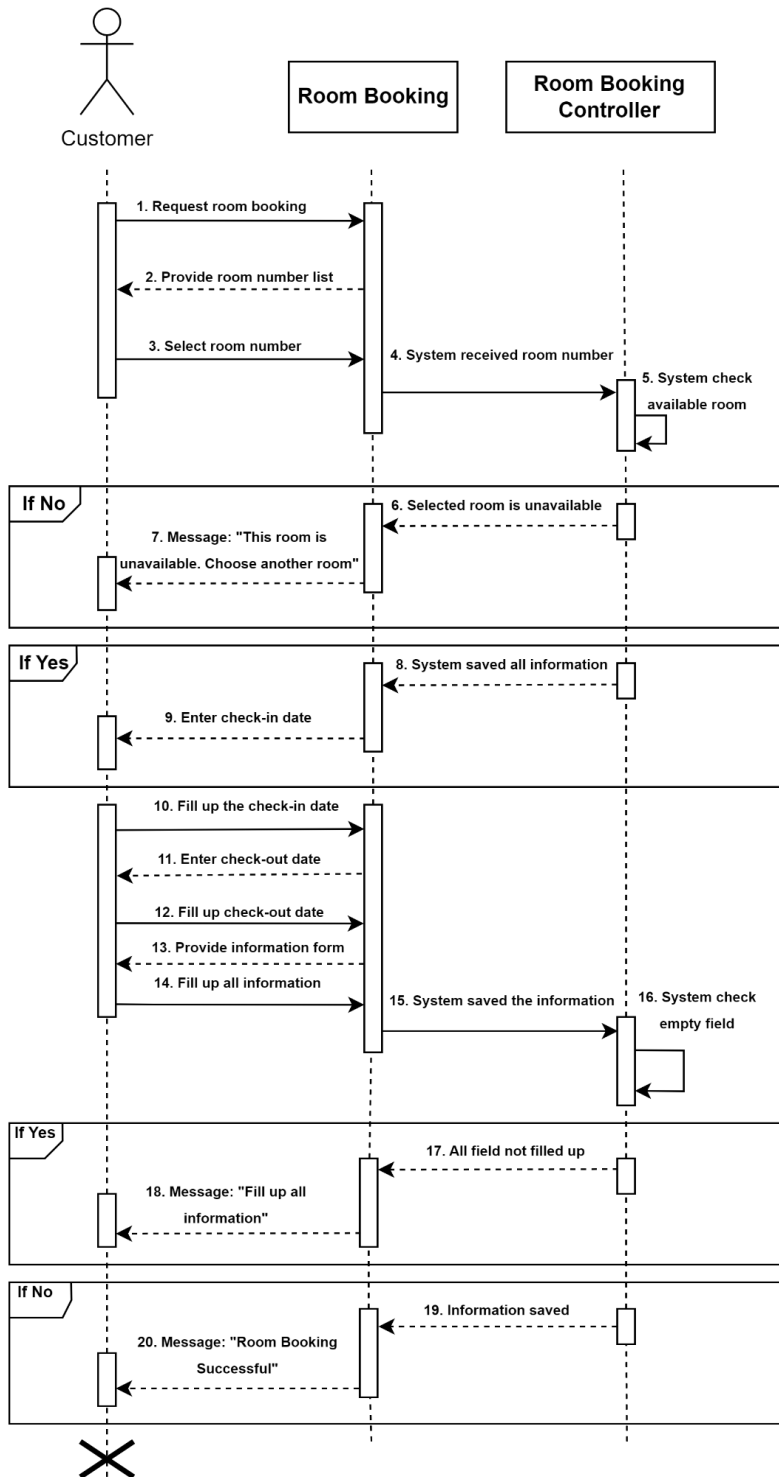


Sequence Diagram

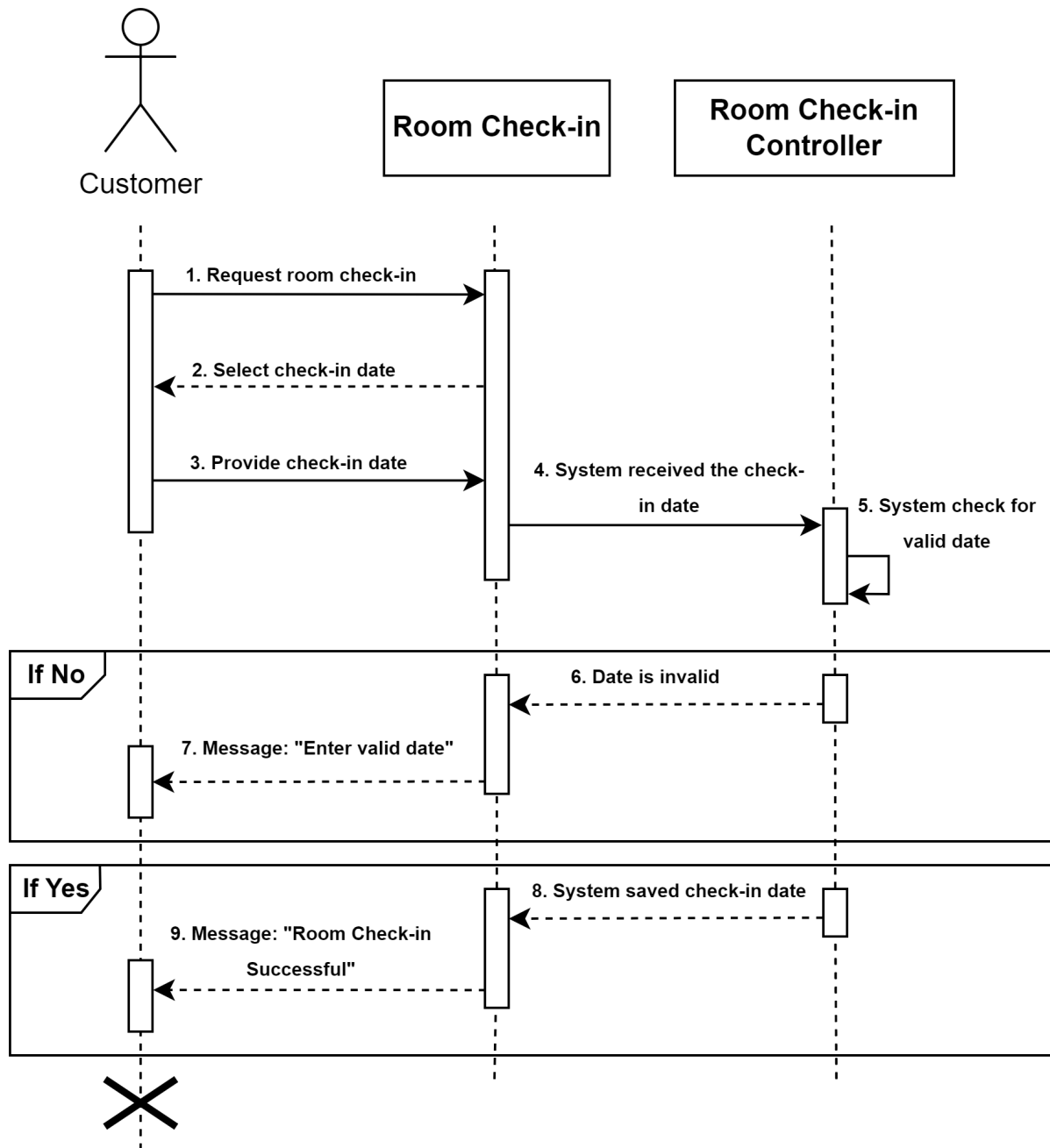
Sequence Diagram - 01: Profile Management



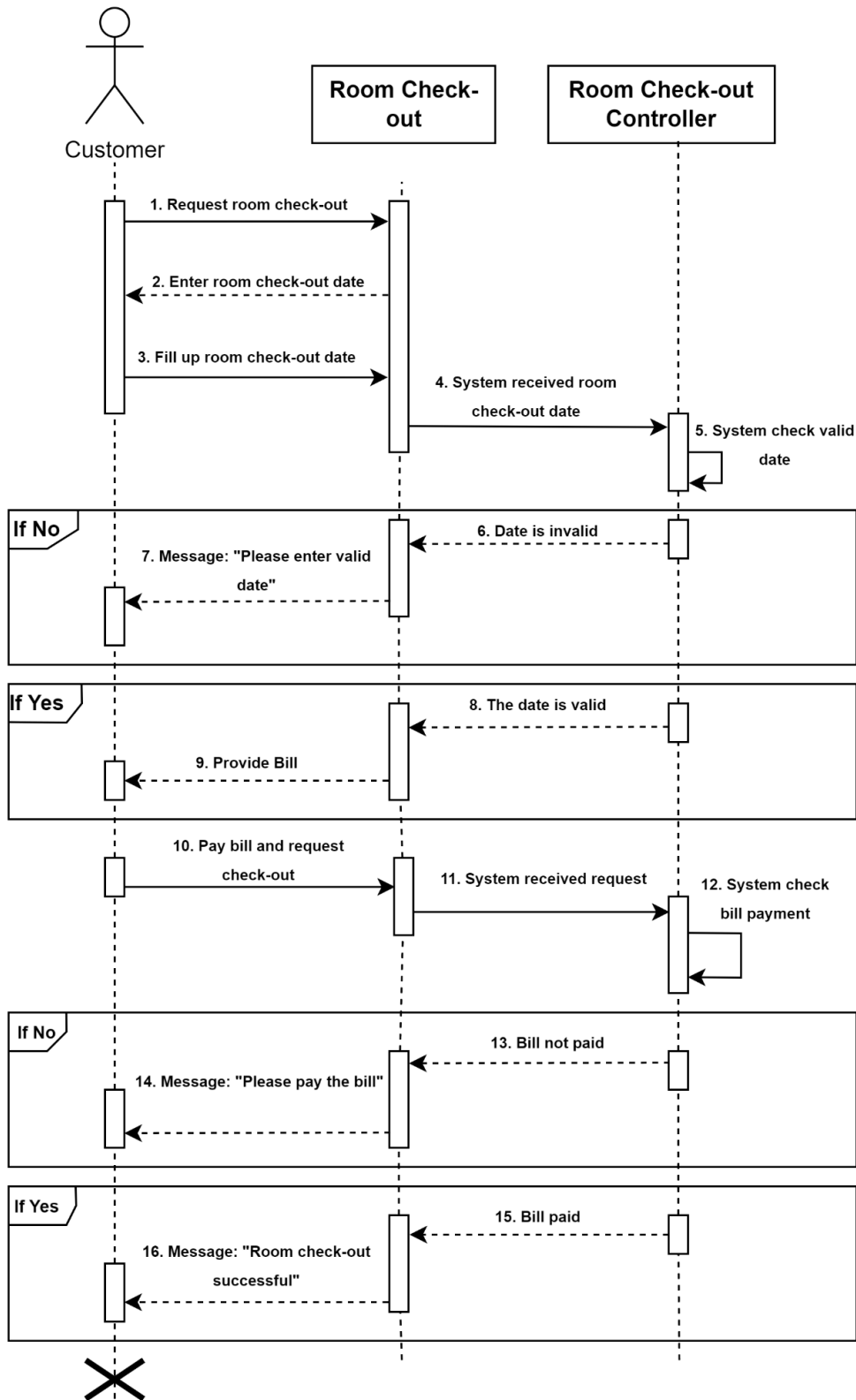
Sequence Diagram - 02: Room Booking



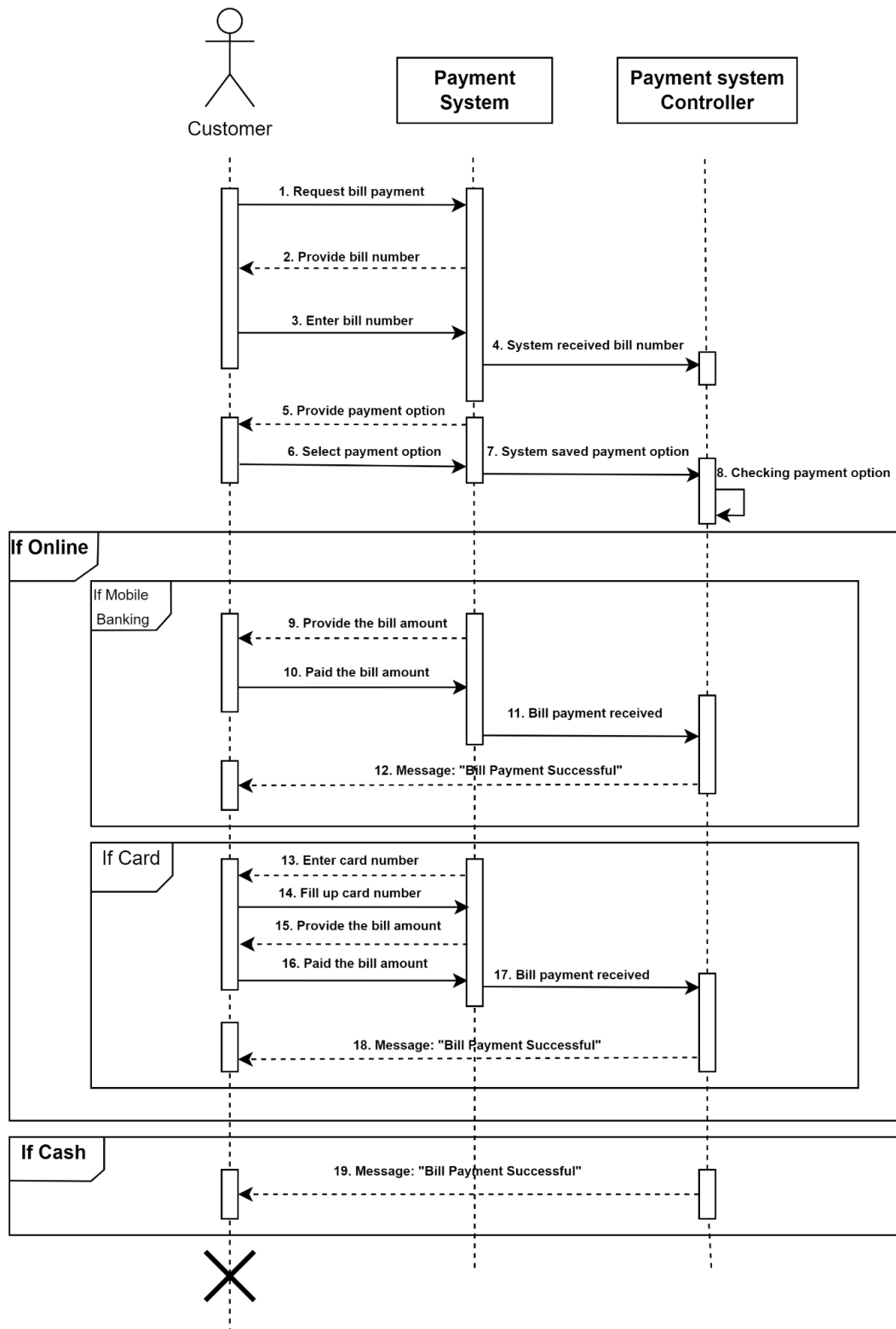
Sequence Diagram - 03: Check-In



Sequence Diagram - 04: Check-out



Sequence Diagram - 05: Payment System



Class Diagram

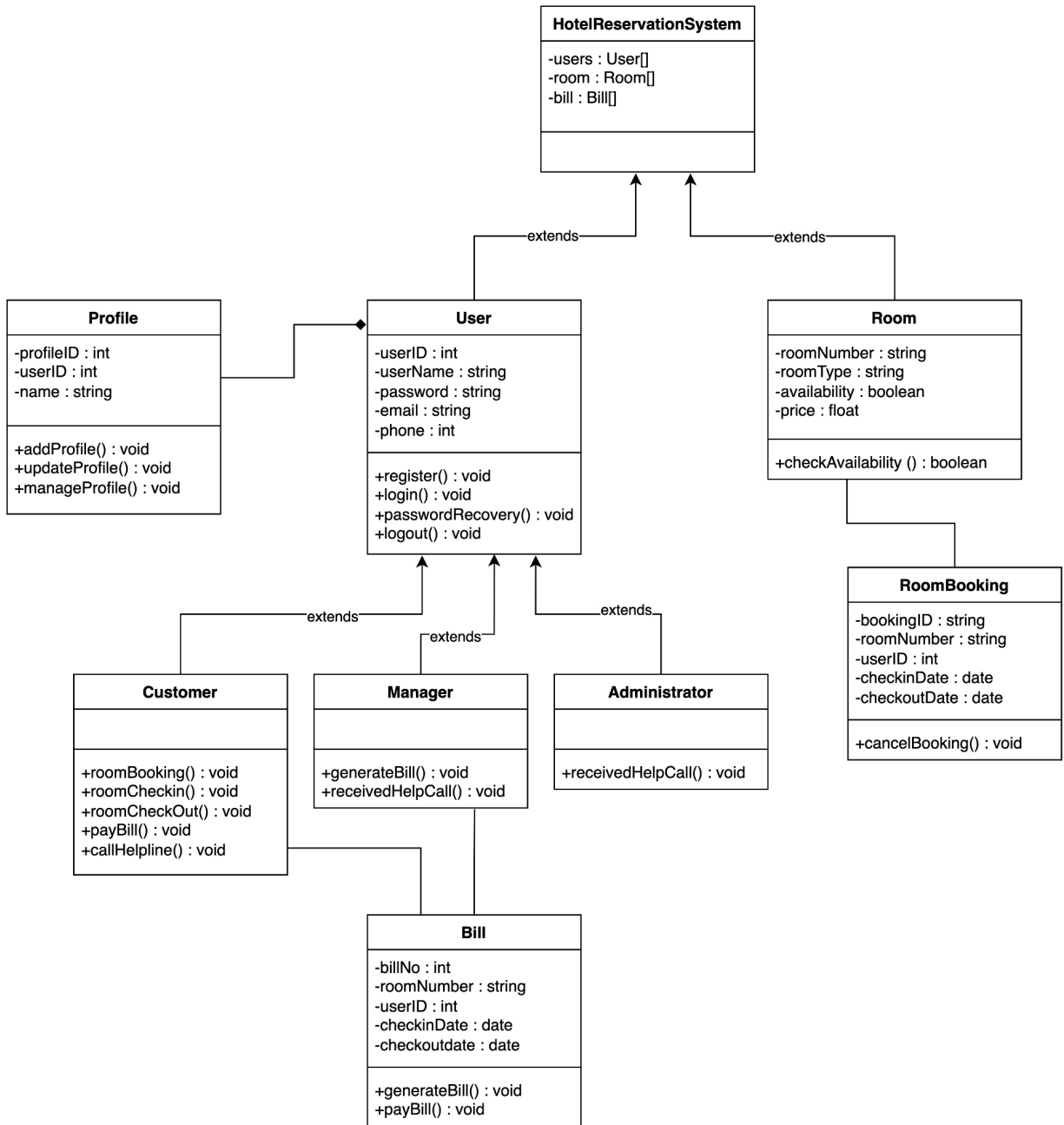


Figure: Class diagram of OTrip

ER Diagram

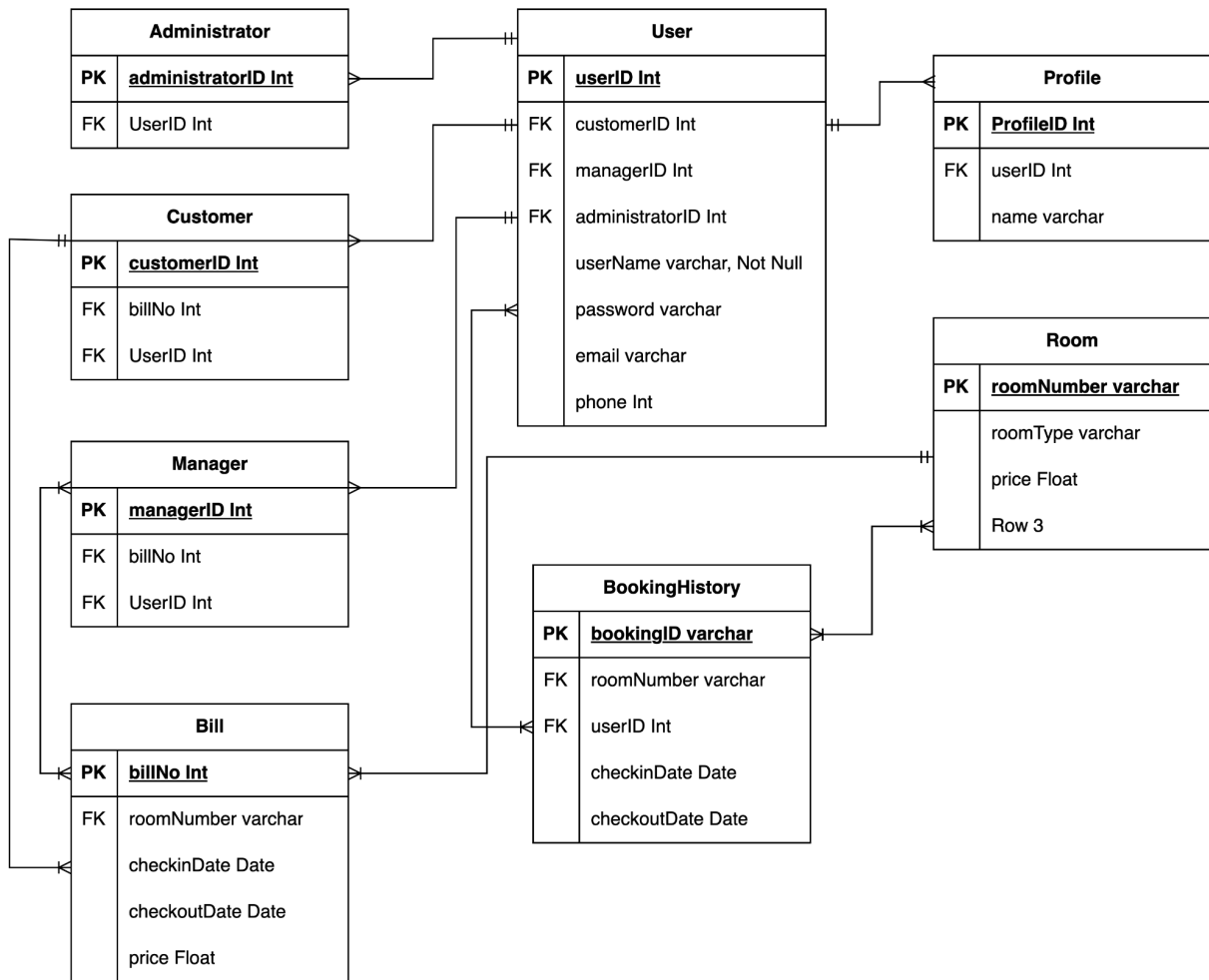


Figure: ER diagram of OTrip